

Turning Simulation into Construction: New Uses of NIZK Simulators

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Abstract. Simulation is a fundamental technique in cryptography, central to security proofs, probably most well-known is its use for showing zero-knowledge. In this paper, we consider simulations no longer being merely a proof technique, but to constructively use simulators of non-interactive zero-knowledge proofs (NIZKs) as an integral component of cryptographic designs. We have occasionally seen such a use, e.g., in designated verifier signatures (DVS), deniable encryption, or the recent concept of anamorphic encryption. However, it was never explicitly explored as a stand-alone concept, which is the goal of this paper.

By embedding simulated proofs directly into concrete systems, we unlock cryptographic functionalities that were previously out of reach under standard assumptions or required prohibitively complex techniques. In other words, by incorporating simulated proofs into the “real” world, rather than the simulated one, we achieve conceptually more elegant primitives. As a primer, we construct a secure signature scheme whose security hinges on a simulated proof of a false statement, i.e., the ludicrous statement $1 = 2$.

To illustrate the broader potential of this approach, we present new and simple constructions of chameleon hash functions with strong privacy guarantees (e.g., full indistinguishability), that do *not* require a trusted setup. Additionally, we present a very simple DVS with tight security proofs and a strengthened notion of non-transferability.

Based on the zero-knowledge guarantees of the underlying NIZKs, the resulting constructions achieve privacy even if the adversary is allowed to choose the random coins to set up the cryptographic material. To model this, we introduce the notion of trapdoor-detectable zero-knowledge, which may be of independent interest.

1 Introduction

Zero-knowledge proofs [13] reveal nothing beyond the truth of a statement. Valid proofs can only be carried out by a prover that knows a witness certifying the truth of the statement. The ingenious idea in [13] was to define the

zero-knowledge property by means of a simulator that can produce valid proofs for statements for which it does not know the witness required to produce a proof. Even more, in non-interactive zero-knowledge (NIZK) proofs [3] the trapdoor associated to the common reference string (CRS) even allows to simulate proofs for false statements. Our question is what if we could use the simulator, typically used to produce proofs of false statements in security proofs, in the real world for good?

In this paper, we bring simulation within NIZK proofs into the real world and demonstrate that even seemingly absurd constructs, like a signature scheme whose “proof” of validity asserts $1 = 2$ can yield meaningful, provably secure schemes. The core idea is elegantly simple: If the only way to forge such a signature is to prove the impossible, then no adversary can succeed — unless they can break the underlying simulation soundness of the proof system.

This mental inversion, i.e., constructing real-world security from simulation, is more than a curiosity. If used carefully, it is a powerful and constructive design principle. As a case study, we revisit chameleon hash functions [19] and designated verifier signatures [17] and show how simulation in the real world enables stronger security guarantees than previously achieved.

Primer: Signatures From $1 = 2$. Before diving into our main constructions and ideas, let us warm up with a simple, yet illuminating, example that captures the essence of the paradigm.

Within NIZK proofs, simulation is based on a trapdoor that is the byproduct of a trusted setup procedure and discarded in the real world. Consequently, while the parameter of this setup, i.e., the so called common reference string (CRS), is available to all parties, the simulation trapdoor is only available to the environment and its only purpose is to serve as a means to proof security. Now the idea is to not use a trapdoor to argue about security, but to achieve it.

To make this idea concrete, we construct a signature scheme whose signing algorithm is nothing more than the simulator of a simulation sound extractable non-interactive zero-knowledge proof system (SSE-NIZK) [14, 26, 27]. The trick is in the choice of the statement being proven, which is defined to be a false statement. Consequently, no honest prover can produce a proof for it, but the simulator can. The signer, holding the simulation trapdoor τ , simply simulates such a proof, bound to the message m being signed.

Thus, the signer proves a statement that is impossible to prove for every message m_i to be signed. Yet, the resulting signature scheme provably achieves EUF-CMA security and in fact, it is even strongly unforgeable, i.e., achieves sEUF-CMA security.⁴ The hardness of forging a signature reduces to the hardness of producing an accepting proof for a false statement. This idea may seem paradoxical, but it is sound, under the sole assumption of simulation-sound extractability.

Contribution. Our central contribution is to demonstrate that when simulation is used carefully and transparently, it can serve as a constructive tool, rather

⁴ One can instead base the construction on a weakly SSE-NIZK [18] and then “only” achieve EUF-CMA security.

than being a mere proof artifact. By explicitly incorporating simulated proofs into the construction of cryptographic primitives, we can realize functionalities that otherwise seem out of reach so far or require significantly stronger assumptions, such as a trusted setup. We showcase this idea through a sequence of constructions and results.

- We begin with a minimalistic *signature scheme* in which each signature is a simulated proof of a false statement, i.e., $1 = 2$. While this may appear paradoxical, we prove that the resulting scheme is strongly unforgeable (sEUF-CMA secure) under the sole assumption that the underlying NIZK proof system is simulation-sound extractable. This toy construction serves as a conceptual primer and illustrates how simulation in the real world can yield meaningful and provably secure functionality.
- As a further application, we construct *designated verifier signature schemes* that are not only conceptually simpler than prior constructions, but also achieve tight security under standard assumptions. Our approach supports enhanced privacy guarantees — such as strong non-transferability and strong unforgeability, without requiring complex constructions. The construction is deliberately minimalistic. So minimal, in fact, that it fits on a beer mat — without compromising formal rigor.
- We then apply the same design principle to construct *chameleon hash functions* with strong privacy guarantees. In particular, a form of indistinguishability that previously required trusted setup. By replacing real proofs with simulated ones in the adaptation interface, we show that strong indistinguishability is achievable without sacrificing full collision-resistance. Additionally, the construction is even conceptually simpler than previous ones.
- We introduce *trapdoor-detectable zero-knowledge proof systems*. They borrow ideas from dual-mode NIZK proof systems, i.e., ones where there is a mode that supports simulation and one that supports extraction. If, however, the randomness used to construct the underlying CRS is given, one can decide in which mode the NIZK is currently operating in. This trick allows to construct primitives where even faced with an adversary that can *choose* the randomness to generate the CRS, one can still retain meaningful privacy guarantees. We showcase this concept for the simulation-sound extractable NIZK proof system due to Groth [14]. This concept might have other future applications and may thus be of independent interest.

Previous and Related Work on Constructive use of Simulations. Simulation is a central tool for cryptographic security proofs (cf. [20] for a comprehensive treatment), dating back to semantic security of encryption [12], the definition of zero-knowledge proofs [13] and secure multi-party computation [10]. In the classical paradigm, a simulator SIM is an algorithm that does not have access to secret inputs but can output fake transcripts or cryptographic objects that are indistinguishable from a real protocol run. Going to the domain of non-interactive zero-knowledge (NIZK) proofs [3] it is well known that in the plain model, e.g., without a common reference string (CRS), such NIZK proof systems can only be designed for trivial languages, i.e., languages in BPP [11]. In a

setting with a CRS where such NIZK proofs can work for any language in NP, the simulator **SIM** is then given a simulation trapdoor τ to be able to compute simulated proofs (even for false statements).

To the best of our knowledge, the first *implicit* use of a simulator in a constructive setting has been used for constructing designated verifier proofs or signatures [17]. In a designated verifier proof Alice wants to prove a statement Ω to Bob in a way that only Bob is convinced of its validity. Jakobsson et al. propose to use Fiat-Shamir transformed Σ -protocols as proofs (or signatures) where in the first message one uses a trapdoor commitment instead as a conventional commitment [17]. If the designated verifier holds the trapdoor to this commitment, they are able to simulate proofs and thus only the verifier (Bob) will be convinced. While implicit in [17], nowadays one typically considers a NIZK proof for a disjunctive statement. More precisely, if Alice wants to convince Bob of the truth of Ω , she provides a proof “ $\Omega \vee$ I know Bob’s secret key”, which to the best of our knowledge was first made explicit in [25]. This interpretation allows to generically instantiate DVS from ring signatures [24]. This can also be interpreted as a simulation by Bob. Around the same time Canetti et al. [4] introduced the concept of deniable encryption. Although not made explicit as a simulator in their work, the so called faking algorithm is an efficient procedure that, given a ciphertext and a chosen fake message, produces alternative randomness (or keys) such that the ciphertext is computationally indistinguishable from a genuine encryption of that fake message. Another example is the MPC in the head (MPCitH) paradigm [16] to construct zero-knowledge proof from MPC protocols. The crucial part is that the prover simulates the run of an MPC protocol “in the head”, commits to the views of all parties and is then challenged by the verifier to open a subset of the parties views.

Finally, the implicit constructive use of a simulator can be found in the recent paradigm of anamorphic encryption (AE) [23]. We recall that AE wants to enable secure communication in the presence of a powerful “dictator”. This entity only allows certain primitives, e.g., public-key encryption (PKE), and knows all the secret keys in a system. The basic idea is to embed a second anamorphic encryption channel into a scheme that allows for transmitting an anamorphic message inside a normal looking ciphertext. This requires so called double keys on the sender and receiver side to encrypt and decrypt anamorphic messages. In [23], the authors suggest to use a PKE scheme following the Naor-Yung (NY) double encryption paradigm [21] to construct AE. We recall that NY transforms an IND-CPA secure PKE scheme into a CCA-secure one. It encrypts the plaintext twice under independent keys and appends a NIZK proof that both ciphertexts encrypt the same message. Normally, it is sufficient to keep a single secret key, but when used in an anamorphic setting the receiver can keep the second secret key (as the decryption double key) and the encryption double key represents the simulation trapdoor of the NIZK proof system. Consequently, one can encrypt a normal message under the first key, an anamorphic message under the second key, and simulate the proof for the now false statement that both messages are the same. The “dictator” will just learn the normal message.

2 Preliminaries

Notation. Let $\lambda \in \mathbb{N}$ denote the security parameter. Unless explicitly stated otherwise, all algorithms implicitly receive 1^λ as auxiliary input. We write $a \leftarrow_{\$} A(x)$ to denote that a is the output of a probabilistic algorithm A on input x . If A is deterministic, we write $a \leftarrow A(x)$. An algorithm is *efficient* if it runs in probabilistic polynomial time (PPT) in λ and in the length of its explicit inputs; unless noted otherwise, all algorithms in this work are assumed to be PPT. If an algorithm A uses external coins, we write $a \leftarrow_{\$} A(x; \xi)$, where ξ denotes its randomness. If output and coins are relevant, we write $(a; \xi) \leftarrow_{\$} A(x)$. Algorithms may output the distinguished error symbol $\perp \notin \{0, 1\}^*$ to indicate failure or an exceptional condition; returning $\perp$ aborts the corresponding algorithm or oracle. We use $(a, \perp) \leftarrow_{\$} A(x)$ to emphasize that a second component is unused.

If S is a finite set, then $a \leftarrow_{\$} S$ means that a is sampled uniformly from S . We denote by $\mathcal{M}$ the message space of a cryptographic scheme, assumed to be efficiently derivable from the scheme’s public parameters or public key. For lists/tuples, we assume an injective, efficiently invertible encoding into $\{0, 1\}^*$.

A function $\nu : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible* if for every polynomial $p(\cdot)$ there exists $n_0 \in \mathbb{N}$ such that for all $n > n_0$, $\nu(n) \leq 1/p(n)$.

Assumptions. We only require one-way functions and the discrete logarithm assumption for most of the constructions and relegate those to Appendix A.

Additional Definitions. We require strongly unforgeable digital signatures schemes (Σ), pseudorandom functions (PRF), and non-interactive commitment-schemes (with checkable public parameters) (Π). Since those are standard, we relegate the formal definitions to Appendix B.

Building Blocks. We now present the building blocks we require. These include a non-interactive commitment scheme with a strengthened hiding notion (Γ), simulation-sound extractable non-interactive zero-knowledge proofs (Π), and chameleon hash functions (CH).

Commitments with Adversarial Setup. We consider a non-interactive commitment scheme with checkable public parameters (cf. Appendix B). We require a notion that we call computational hiding with adversarial setup, which requires that a commitment C does not reveal which message it contains, even if the adversary can choose the message and the public parameters.

Definition 1 (Computational Hiding with Adversarial Setup). A non-interactive commitment scheme $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ is said to be computationally hiding with respect to adversarially generated setup, if for all PPT adversaries $\mathcal{A}$, there exists a negligible function ν such that:

$$\left| \Pr \left[\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{AdvHid}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda),$$

where the experiment $\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{AdvHid}}(\lambda)$ is depicted in Figure 1.

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Exp $\mathcal{A}, \Gamma$ AdvHid( $\lambda$ )
 $b \leftarrow_{\$} \{0, 1\}$ 
 $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{Cmt}_\Gamma}^b}(1^\lambda)$ 
  where oracle  $\mathcal{O}_{\text{Cmt}_\Gamma}^b(\text{pp}_\Gamma, m_0, m_1)$ :
     $(C_0, O_0) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m_0)$ 
     $(C_1, O_1) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m_1)$ 
    return  $\perp$  if  $C_0 = \perp \vee C_1 = \perp$ 
    return  $C_b$ 
  return 1 if  $b^* = b$ 
  return 0

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Fig. 1. Γ Hiding experiment for non-interactive commitment schemes (adversarial pp_Γ)

Instantiation. A suitable construction are Pedersen-Commitments [22], which are binding under the discrete logarithm assumption. To achieve hiding against potentially malicious public parameters, we need to introduce an additional check to verify the well-formedness of the public parameters pp_Γ using CP_Γ . We provide a secure construction in Appendix C.

Designated Verifier Signatures. Designated Verifier Signatures (Ψ) are digital signatures that can be convincingly verified only by a specific recipient, i.e., the designated verifier [17]. To any third party, it is unclear whether the signature was created by the actual signer or forged by the verifier, ensuring non-transferability. This property makes Ψ particularly suitable for scenarios involving sensitive or private communication, where signature forwarding or public verifiability is undesirable.

Framework. There are many variants of DVS such as universal DVS (UDVS) [28] or multi DVS (MDVS), which more recently found increasing interest in Off-the-Record messaging [5]. We now define the syntax for designated verifier signature schemes, formalizing the ideas introduced in [17, 25]. A designated verifier signature scheme ensures that only a specific verifier can check the validity of a signature. For notional convenience, we assume that every public key is an implicit element of the corresponding secret key.

Definition 2 (Designated Verifier Signature Scheme). A designated verifier signature scheme Ψ is a tuple of PPT algorithms

$$\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$$

such that:

PG_Ψ . The setup algorithm takes a security parameter λ and outputs public parameters:

$$\text{pp}_\Psi \leftarrow_{\$} \text{PG}_\Psi(1^\lambda).$$

We assume that pp_Ψ is implicit input to all other algorithms.

KGS_Ψ . The key generation algorithm for the signer takes pp_Ψ and outputs a secret/public key pair:

$$(\text{sk}_\Psi^s, \text{pk}_\Psi^s) \leftarrow_{\$} \text{KGS}_\Psi(\text{pp}_\Psi).$$

KGV_Ψ . The key generation algorithm for the verifier takes pp_Ψ and outputs a secret/public key pair:

$$(\text{sk}_\Psi^v, \text{pk}_\Psi^v) \leftarrow_{\$} \text{KGV}_\Psi(\text{pp}_\Psi).$$

Sign_Ψ . The signing algorithm takes as input a signing key sk_Ψ^s , a verification key pk_Ψ^v , and a message $m \in \{0, 1\}^*$, and outputs a signature:

$$\sigma \leftarrow_{\$} \text{Sign}_\Psi(\text{sk}_\Psi^s, \text{pk}_\Psi^v, m).$$

Verify_Ψ . The verification algorithm takes the verification public key pk_Ψ^v , the public signing key pk_Ψ^s , a message m , and a signature σ , and outputs a decision bit $d \in \{0, 1\}$:

$$d \leftarrow \text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m, \sigma).$$

SIM_Ψ . The simulation algorithm takes the verification secret key sk_Ψ^v , the public signing key pk_Ψ^s , a message m , and outputs a signature σ :

$$\sigma \leftarrow_{\$} \text{SIM}_\Psi(\text{sk}_\Psi^v, \text{pk}_\Psi^s, m).$$

Definition 3 (Correctness). A designated verifier signature scheme is called correct if for all security parameters $\lambda \in \mathbb{N}$, for all $\text{pp}_\Psi \leftarrow_{\$} \text{PG}_\Psi(1^\lambda)$, for all key pairs $(\text{sk}_\Psi^s, \text{pk}_\Psi^s) \leftarrow_{\$} \text{KGS}_\Psi(\text{pp}_\Psi)$, for all $(\text{sk}_\Psi^v, \text{pk}_\Psi^v) \leftarrow_{\$} \text{KGV}_\Psi(\text{pp}_\Psi)$, and all messages $m \in \{0, 1\}^*$, we have:

$$\text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m, \text{Sign}_\Psi(\text{sk}_\Psi^s, \text{pk}_\Psi^v, m)) = 1.$$

and

$$\text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m, \text{SIM}_\Psi(\text{sk}_\Psi^v, \text{pk}_\Psi^s, m)) = 1.$$

Security. We now define the security requirements. In more detail, we use strong unforgeability and simulation-based non-transferability.

Strong Unforgeability. Figure 2 formalizes the strong unforgeability experiment for a designated verifier signature scheme Ψ . The adversary is given access to three oracles: a signing oracle, which returns signatures generated by the honest signer for a honest verifier; a simulation oracle, which produces signatures generated by the verifier using the designated simulator algorithm. Any signature output by the adversary that is valid and has not been previously obtained through any oracle query—neither signing nor simulation—is considered a successful forgery. Strong unforgeability requires that no efficient adversary can produce such a forgery with non-negligible probability, even when having access to both real and simulated signatures, a verification oracle and the possibility to adaptively also get signatures for other keys than the challenge ones. Note, this is actually a stronger definition than one usually uses.

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Exp $\mathcal{A}, \Psi$ DVS-SUnforge( $\lambda$ )
   $\text{pp}_\Psi \leftarrow \text{PG}_\Psi(1^\lambda)$ 
   $(\text{sk}_\Psi^s, \text{pk}_\Psi^s) \leftarrow \text{KGS}_\Psi(\text{pp}_\Psi)$ 
   $(\text{sk}_\Psi^v, \text{pk}_\Psi^v) \leftarrow \text{KGV}_\Psi(\text{pp}_\Psi)$ 
   $\mathcal{Q} \leftarrow \emptyset$ 
   $(m^*, \sigma^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Sign}_\Psi}^{\text{sk}_\Psi^s}, \mathcal{O}_{\text{SIM}_\Psi}^{\text{sk}_\Psi^v}}(\text{pk}_\Psi^s, \text{pk}_\Psi^v)$ 
  where oracle  $\mathcal{O}_{\text{Sign}_\Psi}^{\text{sk}_\Psi^s}(m)$ :
     $\sigma \leftarrow \text{Sign}_\Psi(\text{sk}_\Psi^s, \text{pk}_\Psi^v, m)$ 
    Let  $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$ 
    return  $\sigma$ 
  and where oracle  $\mathcal{O}_{\text{SIM}_\Psi}^{\text{sk}_\Psi^v}(\text{pk}_\Psi^{s'}, m)$ :
     $\sigma \leftarrow \text{SIM}_\Psi(\text{sk}_\Psi^v, \text{pk}_\Psi^{s'}, m)$ 
    If  $\text{pk}_\Psi^{s'} = \text{pk}_\Psi^s$  then  $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$ 
    return  $\sigma$ 
  return 1, if  $\text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m^*, \sigma^*) = 1 \wedge (m^*, \sigma^*) \notin \mathcal{Q}$ 
  return 0

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Fig. 2. Ψ Strong Unforgeability Experiment

Definition 4 (Strong Unforgeability). A designated verifier signature scheme $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ is called *strongly unforgeable* if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:

$$\Pr \left[\text{Exp}_{\mathcal{A}, \Psi}^{\text{DVS-SUnforge}}(\lambda) = 1 \right] \leq \nu(\lambda).$$

Simulation-Based Non-Transferability. Figure 3 defines the experiment for simulation-based non-transferability of a designated verifier signature scheme Ψ . In this experiment, the adversary receives either real signatures or simulated ones via the left-or-right oracle LoRSign , depending on a hidden challenge bit b . Additionally, it is given access to explicit signing and simulation oracles (note, the simulation oracle uses a fixed public key; We later present a stronger experiment, where we drop this restriction). The adversary’s goal is to guess the bit b , i.e., to distinguish between signatures generated by the actual signer and those simulated by the verifier.

Definition 5 (Simulation-based Non-Transferability). A designated verifier signature scheme $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ is said to be *simulation-based non-transferable* if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:

$$\left| \Pr \left[\text{Exp}_{\mathcal{A}, \Psi}^{\text{DVS-SSimNonTransfer}} = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda),$$

Chameleon Hash Functions. Chameleon hash functions (CH) are collision-resistant hash functions equipped with a trapdoor that enables efficient generation of collisions for a fixed input [19]. This controlled malleability makes


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Exp $\mathcal{A}, \Psi$ DVS-SimNonTransfer( $\lambda$ )
 $b \leftarrow_{\$} \{0, 1\}$ 
 $\text{pp}_{\Psi} \leftarrow_{\$} \text{PG}_{\Psi}(1^{\lambda})$ 
 $(\text{sk}_{\Psi}^s, \text{pk}_{\Psi}^s) \leftarrow_{\$} \text{KGS}_{\Psi}(\text{pp}_{\Psi})$ 
 $(\text{sk}_{\Psi}^v, \text{pk}_{\Psi}^v) \leftarrow_{\$} \text{KGV}_{\Psi}(\text{pp}_{\Psi})$ 
 $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{LoRSign}}^{b, \text{sk}_{\Psi}^s, \text{sk}_{\Psi}^v}, \mathcal{O}_{\text{Sign}_{\Psi}}^{\text{sk}_{\Psi}^s}, \mathcal{O}_{\text{SIM}_{\Psi}}^{\text{sk}_{\Psi}^v}}(\text{pk}_{\Psi}^s, \text{pk}_{\Psi}^v)}$ 
  where oracle  $\mathcal{O}_{\text{LoR}}^{b, \text{sk}_{\Psi}^s, \text{sk}_{\Psi}^v}(m)$ :
    If  $b = 0$ :  $\sigma \leftarrow_{\$} \text{Sign}_{\Psi}(\text{sk}_{\Psi}^s, \text{pk}_{\Psi}^v, m)$ 
    If  $b = 1$ :  $\sigma \leftarrow_{\$} \text{SIM}_{\Psi}(\text{sk}_{\Psi}^v, \text{pk}_{\Psi}^s, m)$ 
    Return  $\sigma$ 
  and oracle  $\mathcal{O}_{\text{Sign}_{\Psi}}^{\text{sk}_{\Psi}^s}(\text{pk}_{\Psi}^{v'}, m)$ :
    Return  $\sigma \leftarrow_{\$} \text{Sign}_{\Psi}(\text{sk}_{\Psi}^s, \text{pk}_{\Psi}^{v'}, m)$ 
  and oracle  $\mathcal{O}_{\text{SIM}_{\Psi}}^{\text{sk}_{\Psi}^v}(m)$ :
    Return  $\sigma \leftarrow_{\$} \text{SIM}_{\Psi}(\text{sk}_{\Psi}^v, \text{pk}_{\Psi}^s, m)$ 
  return 1, if  $b^* = b$ 
  return 0

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Fig. 3. Ψ Simulation-Based Non-Transferability

them particularly suited for cryptographic applications where verifiable re-randomization or redaction is required, such as sanitizable signatures or redactable blockchains. Formally, a chameleon hash function behaves like a standard hash to any party without the trapdoor, while the trapdoor holder can find alternative inputs mapping to the same hash output.

Framework. We now present the formal framework for chameleon hash functions. We rely on the definitions given by Derler et al. [7].

Definition 6. A chameleon hash function CH is a tuple of five PPT algorithms $(\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$, such that:

CHPG. The algorithm CHPG , on input a security parameter λ outputs public parameters of the scheme:

$$\text{pp}_{\text{ch}} \leftarrow_{\$} \text{CHPG}(1^{\lambda}).$$

We assume that pp_{ch} is implicit input to all other algorithms.

CHKG. The algorithm CHKG , on input the public parameters pp_{ch} outputs the private and public keys of the scheme:

$$(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}) \leftarrow_{\$} \text{CHKG}(\text{pp}_{\text{ch}}).$$

CHHash. The algorithm CHHash gets as input the public key pk_{ch} , and a message m to hash. It outputs a hash h , and some randomness r .⁵

$$(h, r) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m).$$

⁵ We note that the randomness r is also sometimes called “check value” [1].

CHCheck. The deterministic algorithm CHCheck gets as input the public key pk_{ch} , a message m , randomness r , and a hash h . It outputs a bit $d \in \{0, 1\}$, indicating whether the hash h is valid:

$$d \leftarrow \text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h).$$

CHAdapt. The algorithm CHAdapt on input of a secret key sk_{ch} , the message m , new message m' , randomness r , and hash h outputs new randomness r' , which constitutes a witness of m' under hash h :

$$r' \leftarrow_{\$} \text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r, h).$$

Definition 7 (Correctness). A chameleon hash function is called correct, if for all security parameters $\lambda \in \mathbb{N}$, for all $\text{pp}_{\text{ch}} \leftarrow_{\$} \text{CHPG}(1^\lambda)$, for all $(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}) \leftarrow_{\$} \text{CHKG}(\text{pp}_{\text{ch}})$, for all $m \in \mathcal{M}$, for all $(h, r) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m)$, for all $m' \in \mathcal{M}$, we have for all $r' \leftarrow_{\$} \text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r, h)$, that both $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1$ and $\text{CHCheck}(\text{pk}_{\text{ch}}, m', r', h) = 1$.

Security. We now present the security guarantees we want to achieve.

Full Collision Resistance. The adversary gets access to a collision-finding oracle $\text{CHAdapt}'$, which outputs a collision for the adversarially chosen hash, but also keeps track of each of the queried hash/message pair (h, m) and (h, m') , using the list $\mathcal{Q}$. The adversary wins, if it comes up with a hash/message pair (h^*, m^*) , for the given public key, never queried to or output from the collision-finding oracle.

Now, we formally define full collision-resistance.

Definition 8 (Full Collision-Resistance). A chameleon hash function CH offers full collision-resistance, if for any PPT adversary $\mathcal{A}$ there exists a negligible function ν such that

$$\Pr[\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-CollRes}}(\lambda) = 1] \leq \nu(\lambda),$$

where the corresponding experiment is depicted in Figure 4.

(Normal) Indistinguishability. In a nutshell, (normal) indistinguishability requires that an adversary cannot decide whether randomness was obtained through CHHash or CHAdapt.

Normal Indistinguishability (we sometimes refer to this notion simply as “Indistinguishability”, as this is the standard name in the literature) requires that the randomness r does not reveal if it was obtained through CHHash or CHAdapt.

Upon setup, the challenger generates a key pair $(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}})$ for CH (along with some public parameters pp_{ch}), and draws a bit $b \leftarrow_{\$} \{0, 1\}$. The challenger initializes the adversary with the pk_{ch} and gives the adversary access to a HashOrAdapt oracle, which allows the adversary to submit two messages m, m' . Depending on the bit b , the challenger then either hashes m' directly ($b = 0$),

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Exp $\mathcal{A}, \text{CH}$ F-CollRes( $\lambda$ )
   $\text{pp}_{\text{ch}} \leftarrow_{\$} \text{CHPG}(1^\lambda)$ 
   $(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}) \leftarrow_{\$} \text{CHKG}(\text{pp}_{\text{ch}})$ 
   $\mathcal{Q} \leftarrow \emptyset$ 
   $(m^*, r^*, m'^*, r'^*, h^*) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{CHAdapt}'}^{\text{sk}_{\text{ch}}}}(\text{pk}_{\text{ch}})$ 
  where oracle  $\mathcal{O}_{\text{CHAdapt}'}^{\text{sk}_{\text{ch}}}(m, m', r, h)$ :
    return  $\perp$  if  $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) \neq 1$ 
     $r' \leftarrow_{\$} \text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r, h)$ 
    return  $\perp$  if  $r' = \perp$ 
     $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(h, m), (h, m')\}$ 
    return  $r'$ 
  return 1 if  $\text{CHCheck}(\text{pk}_{\text{ch}}, m^*, r^*, h^*) = 1$  and
     $\text{CHCheck}(\text{pk}_{\text{ch}}, m'^*, r'^*, h^*) = 1$  and
     $m^* \neq m'^*$  and  $(h^*, m^*) \notin \mathcal{Q}$ 
  return 0

```

Fig. 4. CH Full Collision-Resistance.

or first hashes m , and then adapts m to m' ($b = 1$). The resulting hash/randomness pair (h, r) (or (h', r') resp.) is the oracle's output to the adversary. The adversary's objective is to guess the bit b . Note that all keys are generated honestly. The adversary gets access to a collision-finding oracle CHAdapt for arbitrary hashes, meaning that the adversary may also input hashes generated by the HashOrAdapt -oracle.

```

Exp $\mathcal{A}, \text{CH}$ N-Ind( $\lambda$ )
   $\text{pp}_{\text{ch}} \leftarrow_{\$} \text{CHPG}(1^\lambda)$ 
   $(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}) \leftarrow_{\$} \text{CHKG}(\text{pp}_{\text{ch}})$ 
   $b \leftarrow_{\$} \{0, 1\}$ 
   $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{HashOrAdapt}}^{\text{sk}_{\text{ch}}}, \mathcal{O}_{\text{CHAdapt}}^{\text{sk}_{\text{ch}}}}(\text{pk}_{\text{ch}})$ 
  where oracle  $\mathcal{O}_{\text{HashOrAdapt}}^{\text{sk}_{\text{ch}}}(m, m')$ :
     $(h_0, r_0) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m')$ 
     $(h_1, r_1) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m)$ 
     $r'_1 \leftarrow_{\$} \text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r_1, h_1)$ 
    return  $\perp$  if  $r_0 = \perp$  or  $r'_1 = \perp$ 
    return  $(h_0, r_0)$  if  $b = 0$ 
    return  $(h_1, r'_1)$  if  $b = 1$ 
  where oracle  $\mathcal{O}_{\text{CHAdapt}}^{\text{sk}_{\text{ch}}}(m, m', r, h)$ :
    return  $\text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r, h)$ 
  return 1 if  $b^* = b$ 
  return 0

```

Fig. 5. CH (Normal) Indistinguishability With Public Parameters.

Definition 9 ((Normal) Indistinguishability (With Public Parameters)).

A chameleon hash function CH offers (normal) indistinguishability if for any PPT adversary $\mathcal{A}$ there exists a negligible function ν such that

$$\left| \Pr[\mathbf{Exp}_{\mathcal{A}, \text{CH}}^{\text{N-Ind}}(\lambda) = 1] - 1/2 \right| \leq \nu(\lambda).$$

The corresponding experiment is depicted in Figure 5.

3 What can we do with Standard Zero-Knowledge?

We now show what we can do with standard SSE-NIZKs, but before doing so we have to make an important remark.

Remark 1. The simulator underlying the employed SSE-NIZK proof system for the upcoming constructions must neither observe nor program a random oracle, since the construction uses the simulator in the real world without access to any such possibilities.

3.1 Aperitif: Unforgeable Signatures from $1 = 2$

This construction showcases the core idea of our paradigm: real-world simulation as a cryptographic building block.

The main idea is that a signature is a NIZK π on a statement obviously false bound to a message m to be signed. Hence, no valid witness exists, and thus only a simulator can produce verifying proofs. At the core of the construction, the public key is a “forged” crs_{Π} , while the simulation trapdoor τ is used as the secret key. Strong unforgeability directly follows from simulation-sound extractability.

Construction. Let us consider the following NP-language:

$$L_{\Sigma} := \{(m) \mid 1 = 2\} \tag{1}$$

No meaningful witness exists. Thus, only a simulator can produce verifying proofs.

$\text{KG}_{\Sigma}(1^{\lambda})$: Let $(\text{crs}_{\Pi}, \tau) \leftarrow_{\S} \text{SIM}_1(1^{\lambda})$ for L_{Σ} . Return $(\text{sk}, \text{pk}) = (\tau, \text{crs}_{\Pi})$.
 $\text{Sign}_{\Sigma}(\text{sk}, m)$: Parse $\text{sk} = \tau$. Return $\sigma \leftarrow_{\S} \text{SIM}_2(\text{crs}_{\Pi}, \tau, m)$.
 $\text{Verify}_{\Sigma}(\text{pk}, m, \sigma)$: Parse $\text{pk} = \text{crs}_{\Pi}$. Return $\text{Vfy}_{\Pi}(\text{crs}_{\Pi}, m, \sigma)$.

Construction 1: Signature Scheme via Simulation

Security. Subsequently, we prove the security of our signature scheme from Construction 1.

Theorem 1. *Let $\Sigma = (\text{KG}_\Sigma, \text{Sign}_\Sigma, \text{Verify}_\Sigma)$ be the signature scheme from Construction 1. If the NIZK proof system is simulation-sound extractable (SSE), then Σ is strongly unforgeable.*

Intuition. We transform the strong unforgeability experiment step by step. Each hop either switches to an extraction-capable CRS or enforces extraction on any accepting forgery. Any adversary distinguishing the hybrids must break SSE. Thus any accepting fresh proof without an extractable witness contradicts SSE, and forging is infeasible.

The proof is given in Appendix F.

3.2 Signature Digestif: DVS via Constructive Simulation

We conclude our first sweep with designated verifier signatures (DVS). Originally introduced by Jakobsson et al. [17], DVS schemes allow a signer to produce a signature that cannot be transferred to another party. Crucially, the verifier himself can simulate valid-looking signatures, making them non-transferable, i.e., no third party can distinguish between a real signature and one generated by the verifier. This elegantly balances authenticity and deniability.

Our approach directly deploys constructive simulation into the signing and verification process. Thus we obtain a DVS scheme that is not only conceptually cleaner, but also enjoys tight security under standard assumptions. Using the same mind-trick as before, we use the simulation trapdoor τ for generating simulated signatures; The signer simply holds a pre-image w for $y \leftarrow f(w)$, where f is a one-way function. A signature is a proof π such that $y = f(w)$. The verifier's public key is crs_Π .

Construction. The core idea is to fix a one-way function $f : C_\lambda \rightarrow D_\lambda$, and embed knowledge of a preimage as the witness in an NP-statement.

Consider the following NP-language:

$$L_\Psi^\mathcal{F} := \{(f, y, m) \mid \exists(w) : y = f(w)\} \quad (2)$$

and note that the one-way function family $\mathcal{F}$ is already part of the language.

Only the designated verifier — who holds the simulation trapdoor τ — can verify a signature, since the proof, i.e., the signature, is simulatable and therefore not transferable to third parties (in the sense that a verifier can simply generate many more proofs).

We emphasize that the designated verifier publishes the CRS crs_Π used for signing, reversing the traditional trust direction. This inversion is key to our construction and ensures that only the designated verifier, holding the simulation trapdoor τ , can verify signatures, while any third party cannot distinguish them from simulated proofs.

Security. Subsequently, we prove the security of our designated verifier signature scheme from Construction 2.

$\text{PG}_\Psi(1^\lambda)$: Return $\emptyset$.
 $\text{KGS}_\Psi(\text{pp}_\Psi)$: Fix a one-way function $f_\lambda \in \mathcal{F}$, instantiated over domains C_λ and D_λ .
 For brevity, we now use f instead of f_λ . Sample a secret key $w \leftarrow_{\$} C_\lambda$, compute $y \leftarrow f(w)$. Return $(\text{sk}_\Psi^s, \text{pk}_\Psi^s) \leftarrow (w, (f, y))$.
 $\text{KGV}_\Psi(\text{pp}_\Psi)$: Generate a trapdoor CRS:

$$(\text{crs}_\Pi, \tau) \leftarrow_{\$} \text{SIM}_1(1^\lambda) \text{ for } L_\Psi^\mathcal{F}.$$

 Return $(\text{sk}_\Psi^v, \text{pk}_\Psi^v) \leftarrow (\tau, \text{crs}_\Pi)$.
 $\text{Sign}_\Psi(\text{sk}_\Psi^s, \text{pk}_\Psi^v, m)$: Parse $\text{sk}_\Psi^s = w$, $\text{pk}_\Psi^v = \text{crs}_\Pi$, $\text{pk}_\Psi^s = (f, y)$. Let

$$\sigma \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, (f, y, m), w).$$

 Return σ .
 $\text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m, \sigma)$: Parse $\text{pk}_\Psi^v = \text{crs}_\Pi$ and $\text{pk}_\Psi^s = (f, y)$.
 Return $\text{Vfy}_\Pi(\text{crs}_\Pi, (f, y, m), \sigma)$.
 $\text{SIM}_\Psi(\text{sk}_\Psi^v, \text{pk}_\Psi^s, m)$: Parse $\text{sk}_\Psi^v = \tau$, $\text{pk}_\Psi^s = (f, y)$. Return $\text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m))$.

Construction 2: Designated Verifier Signature Scheme Ψ with Parameter $\mathcal{F}$

Theorem 2. *Let $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ be the scheme from Construction 2. Then Ψ is perfectly correct.*

Proof. Follows by construction. $\square$

Theorem 3. *Let $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ be the designated verifier signature scheme from Construction 2. If the NIZK proof system Π is simulation-sound extractable (SSE) and the function family $\mathcal{F}$ is one-way, then Ψ is strongly unforgeable in the sense of Figure 2.*

Intuition. We transform the strong unforgeability experiment step by step. Each hop either switches to an extraction-capable CRS or enforces extraction of a witness for any verifying proof. Any adversary that distinguishes along the way must break SSE. At the end, the extracted witness must satisfy $f(w) = y$, otherwise one-wayness of f is violated.

The proof is given in Appendix G.1

Theorem 4. *Let $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ be the designated verifier signature scheme from Construction 2. If the underlying NIZK proof system is zero-knowledge, then Ψ satisfies simulation-based non-transferability.*

Intuition. Any environment that distinguishes honest signatures from simulated ones must tell apart a real NIZK proof from a simulated one. Each signature contains a NIZK for a disjunction where both branches are computationally indistinguishable by zero-knowledge.

The proof is given in Appendix G.2

4 What If We Strengthen Zero-Knowledge?

We now show what we can achieve, if we strengthen the security guarantees given by the underlying Π . We start by deploying Π s with *composable* zero-knowledge [14].

Informally speaking, composable zero-knowledge says that the receiver of the proof π does not learn anything except the validity of the statement, even if it gets access to the simulation trapdoor τ . Still, the adversary does not have access to the extraction trapdoor ζ or the randomness used.

$$\begin{array}{l}
 \mathbf{Exp}_{\mathcal{A}, \Pi, \text{SIM}}^{\text{Composable-Zero-Knowledge}}(\lambda) \\
 (\text{crs}_\Pi, \tau) \leftarrow_{\$} \text{SIM}_1(1^\lambda) \\
 b \leftarrow_{\$} \{0, 1\} \\
 b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{P_b}(\cdot, \cdot)}(\text{crs}_\Pi, \tau) \\
 \text{where oracle } \mathcal{O}_{P_b}(x, w): \\
 \quad \text{if } R(x, w) \neq 1, \text{ return } \perp \\
 \quad \text{return } \pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, x, w) \\
 \text{and oracle } \mathcal{O}_{P_1}(x, w): \\
 \quad \text{if } R(x, w) \neq 1, \text{ return } \perp \\
 \quad \text{return } \pi \leftarrow_{\$} \text{SIM}_2(\text{crs}_\Pi, \tau, x) \\
 \text{return 1, if } b^* = b \\
 \text{return 0}
 \end{array}$$

Fig. 6. Π Composable Zero-Knowledge

Definition 10 (Composable Zero-Knowledge). A non-interactive proof system Π for language L is composable zero-knowledge, if for any PPT adversary $\mathcal{A}$, there exists an PPT simulator $\text{SIM} = (\text{SIM}_1, \text{SIM}_2)$ such that there exist negligible functions ν_1 and ν_2 such that

$$\left| \Pr [\text{crs}_\Pi \leftarrow_{\$} \text{PG}_\Pi(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] - \Pr [(\text{crs}_\Pi, \tau) \leftarrow_{\$} \text{SIM}_1(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] \right| \leq \nu_1(\lambda),$$

and that

$$\left| \Pr [\mathbf{Exp}_{\mathcal{A}, \Pi, \text{SIM}}^{\text{Composable-Zero-Knowledge}}(\lambda) = 1] - 1/2 \right| \leq \nu_2(\lambda),$$

where the corresponding experiment is depicted in Figure 6.

Suitable instantiations already exist [14, 15].

4.1 Pré-dîner prolongé: Simpler Chameleon-Hashes *With* Trusted Setup

This construction still requires a trusted setup, but continues building up the idea of our line of thought, and may be useful in its own right in certain scenarios. The basic idea remains the same across all different variants we present. In a nutshell, the construction follows the ideas of the presented strongly unforgeable signature scheme from Section 3.1.

Here, crs_Π serves as the public key for the CH, while an adaption requires the simulation trapdoor τ of the Π . To hide m , we use a commitment scheme. The hash value h is simply the commitment, while the check value is a proof that h actually commits to m . For adaption, we use a simulator to prove a false statement that the commitment now opens to a different message m' . As this is (almost always) a false statement, only the simulator can do so.

Unforgeability follows from simulation-sound extractability, while indistinguishability follows from composable zero-knowledge and the hiding definition of the commitment scheme Γ .

Construction. The main idea of our construction is based upon the work done by Derler et al. [6]. Namely, the randomness r is a SSE NIZK attesting membership of a tuple containing the randomness of the commitment C , the hash, as well as the hashed message in the following NP-language:

$$L_{\text{CH}}^\Gamma := \{((C, m, \text{pp}_\Gamma)) \mid \exists(O) : \text{Opn}_\Gamma(\text{pp}_\Gamma, C, O, m) = 1\}, \quad (3)$$

where we will not make pp_Γ explicit henceforth.

This language requires the prover to demonstrate that it knows an opening O such that C is a valid commitment for m . Our construction of a fully collision-resistant CH is presented as Construction 3. We note that compared to Derler et al. [6], we do not rely on a disjunction in the proof. Instead, we only prove one single simple statement.

Security. Subsequently, we prove the security of our CH in Construction 3.

Theorem 5. *Let $\text{CH} = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the construction from Construction 3. If the commitment scheme Γ is binding and the NIZK proof system is simulation-sound extractable (SSE), then CH is fully collision-resistant.*

Intuition. We transform the collision-resistance experiment step by step. First, we switch to an extractable CRS with identical marginal distribution on (crs_Π, τ) . Second, we extract witnesses from *two external* accepting tuples for the same h^* . Two successful extractions yield two distinct openings and break binding; any accepting external proof that resists extraction breaks SSE.

The proof is given in Appendix H.1

Theorem 6. *Let $\text{CH} = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the construction from Construction 3. If the commitment scheme Γ is hiding and the NIZK proof system is composable zero-knowledge (CZK), then CH is (normal) indistinguishable in the sense of Figure 5.*

CHPG(1^λ) : Fix a commitment scheme Γ . Return $\text{pp}_{\text{ch}} = \text{pp}_\Gamma$, where

$$\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda).$$

CHKG(pp_{ch}) : Fix a NIZK scheme for the language L_{CH}^Γ in (3). Let

$$(\text{crs}_\Pi, \tau) \leftarrow_{\$} \Pi.\text{SIM}_1(1^\lambda).$$

Return $(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}) \leftarrow (\tau, \text{crs}_\Pi)$.

CHHash(pk_{ch}, m) : Parse pk_{ch} as crs_Π . Let

$$(C, O) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m), \quad \pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, (C, m), O).$$

Return $(h, r) \leftarrow (C, \pi)$.

CHCheck($\text{pk}_{\text{ch}}, m, r, h$) : Parse pk_{ch} as $(\text{crs}_\Pi, \text{pp}_\Gamma)$, h as C , and r as π . Return 1, if

$$\text{Vfy}_\Pi(\text{crs}_\Pi, (C, m), \pi) = 1,$$

and 0 otherwise.

CHAdapt($\text{sk}_{\text{ch}}, m, m', r, h$) : Parse sk_{ch} as τ , and pk_{ch} as crs_Π .

If $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) \neq 1$, return $\perp$.

Otherwise, return $r' = \pi'$, where

$$\pi' \leftarrow_{\$} \text{SIM}_2(\text{crs}_\Pi, \tau, (C, m')).$$

Construction 3: Our First Construction CH

Intuition. First replace all NIZK proofs produced inside CHHash by simulated ones (CZK). Then fix the message hashed by CHHash to m' independent of the hidden bit b ; commitment hiding makes the view b -independent.

The proof is given in Appendix H.2

4.2 Chameleon-Hashes Without Trusted Setup

We now turn our attention to the construction of chameleon hash functions that provide the notion of strong indistinguishability [8].

Strong Indistinguishability (S-Ind) (Without Trusted Parameters). Strong indistinguishability requires that a randomness r does not reveal whether it was generated using CHHash or CHAdapt, even if no public parameters exist. $\mathcal{A}$ additionally receives the generated secret key. Since the adversary knows the secret key, access to CHAdapt can be simulated by $\mathcal{A}$ internally.

Definition 11 (Strong Indistinguishability (Without Trusted Parameters)). A chameleon hash function CH offers strong indistinguishability without trusted parameters, if for any PPT adversary $\mathcal{A}$ there exists a negligible function ν such that

$$\left| \Pr[\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{S-Ind}}(\lambda) = 1] - 1/2 \right| \leq \nu(\lambda).$$

The corresponding experiments is depicted in Figure 7.

```

Exp $\mathcal{A}, \text{CH}$ S-Ind( $\lambda$ )
  ( $\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}$ )  $\leftarrow_{\$}$  CHKG( $1^\lambda$ )
   $b \leftarrow_{\$} \{0, 1\}$ 
   $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{HashOrAdapt}}^{\text{sk}_{\text{ch}}}}(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}})$ 
  where oracle  $\mathcal{O}_{\text{HashOrAdapt}}^{\text{sk}_{\text{ch}}}(m, m')$ :
    ( $h_0, r_0$ )  $\leftarrow_{\$}$  CHHash( $\text{pk}_{\text{ch}}, m'$ )
    ( $h_1, r_1$ )  $\leftarrow_{\$}$  CHHash( $\text{pk}_{\text{ch}}, m$ )
     $r'_1 \leftarrow_{\$}$  CHAdapt( $\text{sk}_{\text{ch}}, m, m', r_1, h_1$ )
    return  $\perp$  if  $r_0 = \perp$  or  $r'_1 = \perp$ 
    return ( $h_0, r_0$ ) if  $b = 0$ 
    return ( $h_1, r'_1$ ) if  $b = 1$ 
  return 1 if  $b^* = b$ 
  return 0

```

Fig. 7. CH Strong Indistinguishability Without Public Parameters.

Construction: Strong Indistinguishability for CH with Composable Zero-Knowledge We now show that, based on stronger SSE-NIZKs, we can achieve this stronger form of indistinguishability.

The construction paradigm follows the prior construction paradigm, but we now no longer require a trusted setup. Instead also pp_Γ are generated at key-pair generation.

```

CHPG( $1^\lambda$ ): Return  $\emptyset$ .
CHKG( $\text{pp}_{\text{ch}}$ ): Fix a commitment scheme  $\Gamma$ . Fix a NIZK scheme proof system  $\Pi$  for
  language  $L_{\text{CH}}^\Gamma$  in (3). Return  $(\tau, (\text{crs}_\Pi, \text{pp}_\Gamma))$ , where  $\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda)$  and  $(\text{crs}_\Pi, \tau) \leftarrow_{\$} \Pi.\text{SIM}_1(1^\lambda)$ 
CHHash( $\text{pk}_{\text{ch}}, m$ ): Parse  $\text{pk}_{\text{ch}}$  as  $(\text{crs}_\Pi, \text{pp}_\Gamma)$ , and return  $(h, r) = (C, \pi)$ , where
  ( $C, O$ )  $\leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$ , and  $\pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, (\text{pp}_\Gamma, C, m), O)$ .
CHCheck( $\text{pk}_{\text{ch}}, m, r, h$ ): Parse  $\text{pk}_{\text{ch}}$  as  $(\text{crs}_\Pi, \text{pp}_\Gamma)$ ,  $h$  as  $C$ , and  $r$  as  $\pi$ . Return 1, if the
  following holds, and 0 otherwise:
     $\text{Vfy}_\Pi(\text{crs}_\Pi, (\text{pp}_\Gamma, C, m), \pi) = 1$ .
CHAdapt( $\text{sk}_{\text{ch}}, m, m', r, h$ ): Parse  $\text{sk}_{\text{ch}}$  as  $\tau$ . Check that  $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1$ .
  Return  $\perp$ , if not. Otherwise, return  $r' = \pi'$ , where
     $\pi' \leftarrow_{\$} \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m'))$ 

```

Construction 4: Our Second Construction CH'

Security. Subsequently, we prove the security of our CH' in Construction 4.

Theorem 7. Let $\text{CH}' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the scheme from Construction 4. Then CH' is perfectly correct.

Proof. Follows by construction. $\square$

Theorem 8. Let $\text{CH}' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the construction from Construction 4. If the commitment scheme Γ is binding and the NIZK proof system is simulation-sound extractable (SSE), then CH' is fully collision-resistant in the sense of Figure 4.

Intuition. Switch to an extractable CRS with identical marginal on (crs_Π, τ) . Extract from two *external* accepting tuples for the same h^* . Two witnesses break binding; any accept-but-fail on an external tuple breaks SSE.

The proof is given in Appendix I.1

Theorem 9. Let $\text{CH}' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the construction from Construction 4. If Γ is hiding and the NIZK is composable zero-knowledge (CZK), then CH' is strongly indistinguishable (without public parameters) in the sense of Figure 7.

Intuition. Simulate all NIZK proofs produced by CHHash via CZK. Then fix the committed message in $\mathcal{O}_{\text{HashOrAdapt}}$ to m' ; hiding makes the view independent of b .

The proof is given in Appendix I.2

Limitations. We now show that the above construction cannot achieve full-indistinguishability without trusted parameters.

Full Indistinguishability (F-Ind) (Without Public Parameters). Full indistinguishability requires that a randomness r does not reveal whether it was generated using CHHash or CHAdapt, even if the adversary generates all key material, and no public parameters exist. Once more, this also means that the collision-finding oracle can be dropped, as the adversary can find collisions on its own.

Definition 12 (Full Indistinguishability (Without Public Parameters)). A chameleon hash function CH offers full indistinguishability without public parameters, if for any PPT adversary $\mathcal{A}$ there exists a negligible function ν such that

$$\left| \Pr[\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-Ind}}(\lambda) = 1] - 1/2 \right| \leq \nu(\lambda).$$

The corresponding experiments is depicted in Figure 8.

The remaining question is if and how we can achieve full-indistinguishability. The answer is simple: If the adversary can choose the secret key, i.e., $(\text{crs}_\Pi, \tau, \zeta)$, then it always has the extraction trapdoor ζ and can distinguish between honestly generated hashes and simulated ones (because simulated ones cannot be extracted with overwhelming probability). We formalize this with the following theorem:

```

Exp $\mathcal{A}, \text{CH}$ F-Ind( $\lambda$ )
 $b \leftarrow_{\$} \{0, 1\}$ 
 $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{HashOrAdapt}}}(1^\lambda)$ 
  where oracle  $\mathcal{O}_{\text{HashOrAdapt}}(\text{sk}_{\text{ch}}, \text{pk}_{\text{ch}}, m, m')$ :
     $(h_0, r_0) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m')$ 
     $(h_1, r_1) \leftarrow_{\$} \text{CHHash}(\text{pk}_{\text{ch}}, m)$ 
     $r'_1 \leftarrow_{\$} \text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r_1, h_1)$ 
    return  $\perp$  if  $r_0 = \perp$  or  $r'_1 = \perp$ 
    return  $(h_0, r_0)$  if  $b = 0$ 
    return  $(h_1, r'_1)$  if  $b = 1$ 
  return 1 if  $b^* = b$ 
  return 0

```

Fig. 8. CH Full Indistinguishability Without Public Parameters.

Theorem 10. *Let $\text{CH}' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the construction from Construction 4, and let the adversary generate $(\text{crs}_\Pi, \tau, \zeta)$ on its own. Then CH' cannot satisfy Full-Indistinguishability as defined in Figure 8.*

The proof is given in Appendix E.

In a nutshell, the idea is that an adversary can use the extraction trapdoor ζ to distinguish between simulated and honestly generated proofs.

4.3 DVS With Enhanced Non-Transferability

We now show that our construction, if instantiated with a NIZK which offers composable zero-knowledge, achieves the new notion of enhanced simulation-based non-transferability. The idea is as before: The adversary must not be able to distinguish between signatures honestly generated or simulated. However, in the enhanced version, the adversary is given the secret keys, and thus can simulate all (including a non-restricted version of SIM_Ψ), but the LoRSign -oracle (as it uses the hidden bit b), itself.

Definition 13 (Enhanced Simulation-based Non-Transferability). *An enhanced designated verifier signature scheme Ψ is said to be enhanced simulation-based non-transferable if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:*

$$\left| \Pr \left[\mathbf{Exp}_{\mathcal{A}, \Psi}^{\text{DVS-ESimNonTransfer}} = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda),$$

Theorem 11. *Let $\Psi = (\text{PG}_\Psi, \text{KGS}_\Psi, \text{KGV}_\Psi, \text{Sign}_\Psi, \text{Verify}_\Psi, \text{SIM}_\Psi)$ be the scheme from Construction 2. If the underlying NIZK proof system is composable zero-knowledge, then Ψ satisfies enhanced simulation-based non-transferability (Definition 9 / Fig. 9).*

```

Exp $\mathcal{A}, \psi$ DVS-ESimNonDeleg( $\lambda$ )
 $b \leftarrow_{\$} \{0, 1\}$ 
 $(\text{sk}_{\psi}^s, \text{pk}_{\psi}^s) \leftarrow_{\$} \text{KGS}_{\psi}(\text{pp}_{\psi})$ 
 $(\text{sk}_{\psi}^v, \text{pk}_{\psi}^v) \leftarrow_{\$} \text{KGV}_{\psi}(\text{pp}_{\psi})$ 
 $b^* \leftarrow_{\$} \mathcal{A}_{\text{LoR}}^{\mathcal{O}_{\text{LoR}}^{b, \text{sk}_{\psi}^s, \text{sk}_{\psi}^v}}(\text{sk}_{\psi}^s, \text{sk}_{\psi}^v)$ 
  where oracle  $\mathcal{O}_{\text{LoR}}^{b, \text{sk}_{\psi}^s, \text{sk}_{\psi}^v}(m)$ :
    If  $b = 0$ :  $\sigma \leftarrow_{\$} \text{Sign}_{\psi}(\text{sk}_{\psi}^s, \text{pk}_{\psi}^v, m)$ 
    If  $b = 1$ :  $\sigma \leftarrow_{\$} \text{SIM}_{\psi}(\text{sk}_{\psi}^v, \text{pk}_{\psi}^s, m)$ 
    Return  $\sigma$ 
  return 1 if  $b^* = b$ , else 0

```

Fig. 9. Enhanced Simulation-Based Non-Delegatability Experiment

Intuition. Only the LoR oracle depends on b : real proof for $b=0$ vs. simulated proof for $b = 1$. First answer LoR via the CZK challenger’s unified oracle. Then force LoR to always return a simulated proof. Any distinguisher breaks CZK; in the last game the view is b -independent.

The proof is given in Appendix [J.1](#)

5 Beyond Composable Zero-Knowledge

Composable zero-knowledge has become a standard notion for proving security of cryptographic protocols in complex environments, particularly in simulation-based frameworks. However, the requirements of composability inherently preclude certain optimizations in protocol design. In this section, we depart from the composability paradigm and investigate an alternative notion, i.e., *trapdoor-detectable zero-knowledge*.

We show that this definition permits the construction of primitives with enhanced privacy properties, such as full simulation-based non-transferability, even in adversarially controlled key environments. The core idea is to always embed simulation trapdoors, but make the use of an extraction trapdoor detectable for the holder of the randomness used to generate the corresponding crs_{Π} .

5.1 Trapdoor-Detectable Zero-Knowledge

Trapdoor-detectable zero-knowledge (TD-ZK) augments standard NIZK systems with a predicate that determines whether a given common reference string (CRS) contains an embedded extraction trapdoor, if the randomness used to generate crs_{Π} is known. More precisely, the prover algorithm remains unchanged, but the setup algorithm is augmented to output auxiliary trapdoor data, and the system provides a predicate DtctTd_{Π} that detects whether this data was used to generate the CRS. Thus, the CRS can either be set up with $b = 0$, i.e., no extraction trapdoor, or $b = 1$ which allows for extraction.

The key property of TD-ZK is that the detection predicate DtctTd_Π allows honest parties to ensure that a CRS used was generated in a trapdoor-free manner (i.e., with $b = 0$), while enabling simulation in the real world for CRS generated with $b = 1$ and known trapdoor seed ξ . This structural feature enables simulation-based security models to coexist with real-world implementations in which no extraction trapdoor is embedded.

In the remainder of this work, we employ TD-ZK to construct primitives that admit simulation in the real world without trusted setup assumptions. The central insight is that such constructions do not require extraction trapdoors, but only the ability to simulate proofs when a specific CRS and seed are used together.

Definition 14 (Non-Interactive Proof System with Trapdoor Detection). *A non-interactive proof system Π for an NP language L with trapdoor detection consists of the following PPT algorithms:*

$$(\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$$

such that:

PG_Π . *The parameter generation algorithm takes a security parameter λ and a trapdoor bit $b \in \{0, 1\}$ and outputs a common reference string crs_Π and associated randomness ξ :*

$$(\text{crs}_\Pi; \xi) \leftarrow_{\$} \text{PG}_\Pi(1^\lambda, b)$$

If $b = 1$, then ξ can be used to extract an extraction trapdoor ζ . Else, i.e., $b = 0$, then crs_Π is extraction trapdoor-free.

DtctTd_Π . *The trapdoor detection algorithm takes the randomness ξ , and returns a bit $d \in \{0, 1\}$ indicating whether the CRS was generated with an extraction trapdoor:*

$$d \leftarrow \text{DtctTd}_\Pi(\text{crs}_\Pi, \xi)$$

Prf_Π . *The prover takes the CRS crs_Π , a statement $x \in L$, and a witness w with $R(x, w) = 1$, and returns a proof:*

$$\pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, x, w)$$

Vfy_Π . *The verifier takes the CRS crs_Π , a statement x , and a proof π , and returns a decision bit $d \in \{0, 1\}$:*

$$d \leftarrow \text{Vfy}_\Pi(\text{crs}_\Pi, x, \pi)$$

Definition 15 (Correctness). *A non-interactive proof system with trapdoor detection is called correct, if for all $\lambda \in \mathbb{N}$, for all $b \in \{0, 1\}$, for all $(\text{crs}_\Pi; \xi) \leftarrow_{\$} \text{PG}_\Pi(1^\lambda, b)$, for all $x \in L$, for all w such that $R(x, w) = 1$, for all $\pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, x, w)$, it holds that:*

$$\text{Vfy}_\Pi(\text{crs}_\Pi, x, \pi) = 1 \quad \text{and} \quad \text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) = (b = 1).$$

Security. We now formalize the security properties of TD-ZK.

Definition 16 (CRS-Indistinguishability). Let $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$ be a non-interactive proof system with trapdoor detection.

We say that Π satisfies CRS-Indistinguishability if there exists a negligible function $\nu(\lambda)$ and a PPT simulator SIM_1 , such that for all PPT adversaries $\mathcal{A}$,

$$\left| \Pr \left[\begin{array}{l} (\text{crs}_\Pi; \xi) \leftarrow_{\$} \text{PG}_\Pi(1^\lambda, 0); \\ (\tau, \perp) \leftarrow_{\$} \text{SIM}_1(\xi, 0); \\ \mathcal{A}(\text{crs}_\Pi, \tau) = 1 \end{array} \right] - \Pr \left[\begin{array}{l} (\text{crs}_\Pi; \xi) \leftarrow_{\$} \text{PG}_\Pi(1^\lambda, 1); \\ (\tau, \zeta) \leftarrow_{\$} \text{SIM}_1(\xi, 1); \\ \mathcal{A}(\text{crs}_\Pi, \tau) = 1 \end{array} \right] \right| \leq \nu(\lambda).$$

Exp _{$\mathcal{A}, \Pi, \text{SIM}$} ^{TD-ZK}(λ)

$b \leftarrow_{\$} \{0, 1\}$

$b^* \leftarrow \mathcal{A}^{\mathcal{O}_{\text{POK}}^b}(1^\lambda)$

where oracle $\mathcal{O}_{\text{POK}}^b(\text{crs}_\Pi, \xi, x, w)$:

if $R(x, w) = 0$ or $\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) \neq 0$: return $\perp$

let $(\tau, \perp) \leftarrow_{\$} \text{SIM}_1(\xi, 0)$

if $b = 0$: return $\text{Prf}_\Pi(\text{crs}_\Pi, x, w)$

if $b = 1$: return $\text{SIM}_2(\text{crs}_\Pi, \tau, x)$

return 1 if $b^* = b$

return 0

Fig. 10. Trapdoor-ZK Proof Indistinguishability Experiment

Definition 17 (Trapdoor-Detectable Zero-Knowledge). A non-interactive proof system with trapdoor detection Π satisfies trapdoor-zero-knowledge, if there exists a simulator $\text{SIM} = (\text{SIM}_1, \text{SIM}_2)$, see below for the interfaces, such that for all PPT adversaries $\mathcal{A}$,

$$\left| \Pr \left[\text{Exp}_{\mathcal{A}, \Pi, \text{SIM}}^{\text{TD-ZK}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda)$$

The experiment is given in Figure 10.

In contrast to standard zero-knowledge simulators, we need to slightly adjust the simulators' interfaces:

- $\text{SIM}_1(\xi, b)$ returns a pair (τ, ζ) , where:
 - τ is a simulation trapdoor, and
 - ζ is an extraction trapdoor (only present if $b = 1$; Otherwise we obtain $\zeta = \perp$).

- $\text{SIM}_2(\text{crs}_\Pi, \tau, x)$ returns a simulated proof π such that

$$\text{Vfy}_\Pi(\text{crs}_\Pi, x, \pi) = 1$$

even for wrong statements.

Modified Simulation-Sound Extractability. Due to the changed interfaces, we need to slightly alter the standard SSE experiment.

```

Exp $\mathcal{A}, \Pi, \mathcal{E}$ MSimSoundExt( $\lambda$ )
  ( $\text{crs}_\Pi; \xi$ )  $\leftarrow_{\$}$   $\text{PG}_\Pi(1^\lambda, 1)$ 
  ( $\tau, \zeta$ )  $\leftarrow_{\$}$   $\text{SIM}_1(\xi, 1)$ 
   $\mathcal{Q} \leftarrow \emptyset$ 
  ( $x^*, \pi^*$ )  $\leftarrow_{\$}$   $\mathcal{A}^{\mathcal{O}_{\text{SIM}}^\tau(\cdot)}(\text{crs}_\Pi)$ 
  where oracle  $\mathcal{O}_{\text{SIM}}^\tau(x)$ :
    obtain  $\pi \leftarrow_{\$}$   $\text{SIM}_2(\text{crs}_\Pi, \tau, x)$ 
     $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(x, \pi)\}$ 
  return  $\pi$ 
   $w^* \leftarrow_{\$}$   $\mathcal{E}(\text{crs}_\Pi, \zeta, x^*, \pi^*)$ 
  return 1, if  $\text{Vfy}_\Pi(\text{crs}_\Pi, x^*, \pi^*) = 1 \wedge R(x^*, w^*) = 0 \wedge (x^*, \pi^*) \notin \mathcal{Q}$ 
  return 0

```

Fig. 11. Π Modified Simulation Sound Extractability

Definition 18 (Modified Simulation-Sound Extractability). A non-interactive zero-knowledge proof system with trapdoor detection Π for a language L is said to be modified simulation-sound extractable, if for every PPT adversary $\mathcal{A}$ there exists a PPT extractor $\mathcal{E}$, such that for all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr \left[\text{Exp}_{\mathcal{A}, \Pi, \mathcal{E}}^{\text{MSimSoundExt}}(\lambda) = 1 \right] \leq \nu(\lambda),$$

where the experiment is defined in Figure 11, and $\text{SIM} = (\text{SIM}_1, \text{SIM}_2)$ is the simulator pair from Definition 17.

5.2 Trapdoor-Detectable SSE NIZK Instantiation

We now show how to construct a trapdoor-detectable SSE NIZK. In particular, we show how one can slightly rephrase the scheme by Groth [14].

In the original paper, Groth combines a commitment-scheme based on DLIN with an IND-CPA-secure encryption scheme. In the “trapdoor”-settings, Groth keeps the secret key for the encryption scheme and initializes the commitment-scheme in a hiding setting. As our setting lies “in-between”, simulation must always be possible, while decryption must *not* be possible. We do so by altering

the generation of the CRS. See Construction 5, where $\mathcal{H}_{\mathbb{G}_1} : \{0, 1\}^* \rightarrow \mathbb{G}_1$ is a random oracle. Also, let $\mathcal{H}_\xi : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$. Let $\text{PRFEval} : \{0, 1\}^\lambda \rightarrow \mathbb{Z}_p$ be a pseudo-random function, and $\text{PRG} : \{0, 1\}^\lambda \rightarrow (\{0, 1\}^\lambda \times \{0, 1\}^\lambda)$ be a pseudorandom generator.

Fixed Parameters. Fix a bilinear group $(\mathbb{G}_1, \mathbb{G}_T, e, p, g_1, g_T) \leftarrow_{\$} \text{BGGen}(1^\lambda)$. All algorithms are implicitly parameterized by the bilinear group. We write $\text{PG}_\Pi(1^\lambda, b; \xi)$, $\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi)$, $\text{SIM}_1(\xi, b)$ with the bilinear group implicit. All randomness and outputs are deterministic given ξ , the bilinear group, and the random oracles.

The construction is given in Construction 5.

$\text{PG}_\Pi(1^\lambda, b) :$

- Draw $\xi \leftarrow_{\$} \{0, 1\}^\lambda$ (if randomness is not provided),
- Let $\xi' \leftarrow \mathcal{H}_\xi(\xi)$,
- Generate $(\text{sk}_\Sigma, \text{pk}_\Sigma) \leftarrow_{\$} \text{KG}_\Sigma(1^\lambda; \xi')$,
- If $b = 0$:

$$f \leftarrow \mathcal{H}_{\mathbb{G}_1}(1, \xi'), h \leftarrow \mathcal{H}_{\mathbb{G}_1}(2, \xi')$$
- If $b = 1$:

$$x \leftarrow \text{PRFEval}(\xi', 1), y \leftarrow \text{PRFEval}(\xi', 2), f \leftarrow g_1^x, h \leftarrow g_1^y$$
- $r_u \leftarrow \text{PRFEval}(\xi', 3), s_v \leftarrow \text{PRFEval}(\xi', 4)$. Let $u \leftarrow f^{r_u}, v \leftarrow h^{s_v}, w \leftarrow g_1^{r_u + s_v}$.
- Let $\text{crs}_\Pi \leftarrow (f, h, u, v, w, \text{pk}_\Sigma)$
- Return $(\text{crs}_\Pi; \xi)$.

$\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) :$

- Let $\text{crs}'_\Pi \leftarrow \text{PG}_\Pi(1^\lambda, 0; \xi)$
- Let $\text{crs}''_\Pi \leftarrow \text{PG}_\Pi(1^\lambda, 1; \xi)$
- Return $\perp$ if $\text{crs}''_\Pi = \text{crs}'_\Pi$
- Return 1 if $\text{crs}''_\Pi = \text{crs}_\Pi$
- Return 0 if $\text{crs}'_\Pi = \text{crs}_\Pi$
- Return $\perp$

$\text{SIM}_1(\xi, b) :$

- If $b = 0$, run $\text{PG}_\Pi(1^\lambda, 0; \xi)$, but keep $(\tau, \zeta) = ((r_u, s_v, \text{sk}_\Sigma), \perp)$.
- If $b = 1$, run $\text{PG}_\Pi(1^\lambda, 1; \xi)$, but keep $(\tau, \zeta) = ((r_u, s_v, \text{sk}_\Sigma), (x, y))$.
- Return (τ, ζ)

Construction 5: Altered Algorithms for Groth (TD-ZK compatible). The proving, verifying and the simulator SIM_2 algorithms are unchanged. The signature scheme used is the one by Groth [14].

Security. Subsequently, we prove the security of our TD-ZK Π in Construction 5.

Theorem 12. *Let $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$ be the scheme from Construction 5. Then Π is correct (with a negligible error).*

Intuition. Proving and verification are identical to Groth’s scheme. Only the CRS generation changes. A verifier can reject a true statement only if a rare CRS pathology occurs: the $b \in \{0, 1\}$ branches collide into the same (f, h) . This event is negligible in λ . Otherwise, completeness is perfect.

The proof is given in Appendix M.1.

Theorem 13. *Let $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$ be the scheme from Construction 5. Then Π is CRS-indistinguishable.*

Intuition. Mode $b=0$: (f, h) are independent RO images in $\mathbb{G}_1$; mode $b=1$: $(f, h) = (g_1^x, g_1^y)$ with PRF-derived exponents in $\mathbb{Z}_p$. Both yield (computationally) uniform (f, h) in $\mathbb{G}_1$. The remaining CRS parts and the simulator handle τ as the same deterministic functions of the seed ξ' in both modes. Thus (crs_Π, τ) is indistinguishable across $b \in \{0, 1\}$.

The proof is given in Appendix M.2.

Theorem 14. *Let $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$ be the scheme from Construction 5. Then Π satisfies trapdoor-detectable zero-knowledge in the sense of Figure 10.*

Intuition. The oracle serves answers only on admissible queries: $R(x, w) = 1$ and $\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) = 0$. For such queries, branch $b = 0$ returns real proofs, while branch $b = 1$ returns simulated proofs using the *same* seed-derived simulation trapdoor $\tau \leftarrow \text{SIM}_1(\xi, 0)$. Hybridize answers one-by-one from real to simulated; each swap is hidden by the ZK of the underlying proof system.

Theorem 15. *Let $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$ be the scheme from Construction 5. If the underlying NIZK is simulation-sound extractable (SSE) in the standard sense, then Π satisfies Modified Simulation-Sound Extractability in the sense of Figure 11.*

Proof. The modified experiment is isomorphic to the standard SSE game via identical distribution of the CRS, while Prf_Π and Vfy_Π are not altered.

5.3 The Final Form of Our Chameleon-Hash CH''

We now present our final construction. It still follows the original paradigm. However, instead of deriving a secret key directly, we store the randomness ξ to derive all secret keys. We can then “force” compliant behavior, even if the randomness ξ is chosen by the adversary $\mathcal{A}$, as we can check that crs_Π is generated honestly, i.e., without an extraction trapdoor ζ .

The main trick lies in the proof: In the proof, we can generate ζ , while in the non-simulated case, ζ does not exist, regardless of the randomness (ignoring some negligible probability). As our zero-knowledge definition allows for giving away the simulation trapdoor τ , this construction achieves full-indistinguishability even if $\mathcal{A}$ can chose the randomness at key-generation.

$\text{CHPG}(1^\lambda)$: Return $\emptyset$.
 $\text{CHKG}(\text{pp}_{\text{ch}})$: Fix a commitment scheme Γ .
 Fix a trapdoor-detectable non-interactive proof system
 $\Pi = (\text{PG}_\Pi, \text{DtctTd}_\Pi, \text{Prf}_\Pi, \text{Vfy}_\Pi)$
 for language L_{CH}^Γ from (3). Sample:
 $(\text{pp}_\Gamma; \xi_\Gamma) \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda), \quad (\text{crs}_\Pi; \xi_\Pi) \leftarrow_{\$} \text{PG}_\Pi(1^\lambda, b = 0)$.
 Return:
 $\text{sk}_{\text{ch}} \leftarrow (\xi_\Gamma, \xi_\Pi), \quad \text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$.
 $\text{CHHash}(\text{pk}_{\text{ch}}, m)$: Parse pk_{ch} as $(\text{crs}_\Pi, \text{pp}_\Gamma)$. Sample:
 $(C, O) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m), \quad \pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, (\text{pp}_\Gamma, C, m), O)$.
 Return $(h, r) \leftarrow (C, \pi)$.
 $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h)$: Parse $\text{pk}_{\text{ch}} = (\text{crs}_\Pi, \text{pp}_\Gamma)$, $h = C$, $r = \pi$. Return 1 if:
 $\text{Vfy}_\Pi(\text{crs}_\Pi, (\text{pp}_\Gamma, C, m), \pi) = 1$.
 $\text{CHAdapt}(\text{sk}_{\text{ch}}, m, m', r, h)$: Parse $\text{sk}_{\text{ch}} = (\xi_\Gamma, \xi_\Pi)$, $\text{pk}_{\text{ch}} = (\text{crs}_\Pi, \text{pp}_\Gamma)$, $r = \pi$, $h = C$.
 Let $\text{pp}'_\Gamma \leftarrow \text{PG}_\Gamma(\xi_\Gamma)$.
 Return $\perp$ if $\text{pp}'_\Gamma \neq \text{pp}_\Gamma$
 Check that:
 $\text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1 \quad \text{and} \quad \text{DtctTd}_\Pi(\text{crs}_\Pi, \xi_\Pi) = 0$.
 If not, return $\perp$.
 Let $(\tau, \perp) \leftarrow_{\$} \text{SIM}_1(\xi_\Pi)$.
 Then compute:
 $\pi' \leftarrow_{\$} \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m'))$
 Check that:
 $\text{CHCheck}(\text{pk}_{\text{ch}}, m', \pi', h) = 1$
 If not, return $\perp$.
 Return $r' = \pi'$.

Construction 6: Final Construction: CH'' with Full-Indistinguishability Without Trusted Parameter Generation

Security. Now we prove the security of our chameleon-hash CH'' with full-indistinguishability without trusted parameter generation from Construction 6.

Theorem 16. *Let $\text{CH}'' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the scheme from Construction 6. Then CH'' is perfectly correct.*

Proof. Follows by construction. $\square$

Theorem 17 (Full-Collision-Resistance of CH''). *Let $\text{CH}'' = (\text{CHPG}, \text{CHKG}, \text{CHHash}, \text{CHCheck}, \text{CHAdapt})$ be the scheme from Construction 6. If the com-*

mitment scheme Γ is binding and the non-interactive proof system Π is CRS-indistinguishable and modified simulation-sound extractable, then CH'' satisfies Full-Collision-Resistance as in Fig. 4.

Intuition. Switch to an extractable CRS via CRS-indistinguishability. Log external vs. simulated tuples. Extract from two external accepting tuples for the same h^* . Two successes break binding; any accept-but-fail on an external tuple breaks modified SSE.

The proof is given in Appendix K.1

Theorem 18. *Let CH'' be the scheme from Construction 6. If the NIZK proof system provides trapdoor/composable zero-knowledge and the commitment scheme Γ is adversarial-hiding, then CH'' satisfies Full-Indistinguishability as in Fig. 8.*

Intuition. First simulate the NIZK proof inside CHHash using the trapdoor. Then in the challenge oracle always commit to m' ; with simulated proofs, commitment hiding removes any dependence on b .

The proof is given in Appendix K.2.

5.4 The Final Form of Our DVS Ψ''

We now present our final DVS construction, denoted by Ψ'' , which replaces the composable zero-knowledge component with a trapdoor-detectable zero-knowledge (TD-ZK) proof system. The core advantage of Ψ'' lies in its ability to achieve enhanced privacy guarantees, captured by the notion of full simulation-based non-transferability defined below.

Informally speaking, full non-transferability now also holds when the adversary is permitted to choose the randomness used during key generation—an important strengthening over prior notions.

In the following, we first define the enhanced privacy experiment formally, detail our construction, and subsequently analyze correctness, strong unforgeability, and privacy in a rigorous manner.

Full Simulation-Based Non-Transferability. We now formally define *full simulation-based non-transferability* for a designated verifier signature scheme Ψ'' . This notion significantly strengthens previous definitions of non-transferability by granting the adversary $\mathcal{A}$ complete control over the secret keys. More precisely, in the following experiment, the adversary not only attempts to distinguish between real and simulated signatures, but is also permitted to adversarially generate or choose both the signer's secret key sk_{Ψ}^s and the verifier's secret key sk_{Ψ}^v , as well as the corresponding public keys. Thus, the adversary is explicitly allowed to construct maliciously structured keys (as long as they remain valid outputs of the respective key-generation algorithms), aiming at facilitating the distinguishability of signatures.

Consequently, the resulting notion ensures indistinguishability between genuinely created signatures and simulated ones *even if the adversary controls the*

keys used. This property provides robust protection against key-related adversarial strategies and captures the intuitive notion of non-transferability in the presence of potentially maliciously generated parameters. We formalize this strong notion of non-transferability in Figure 12.

```

Exp $\mathcal{A}, \Psi''$ DVS-FSimNonTransfer( $\lambda$ )
   $b \leftarrow_{\$} \{0, 1\}$ 
   $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{LoRSign}}^b}(1^\lambda)$ 
  where oracle  $\mathcal{O}_{\text{LoR}}^b(\text{sk}_\psi^s, \text{sk}_\psi^v, m)$ :
     $\sigma_0 \leftarrow_{\$} \Psi'' \text{sign}(\text{sk}_\psi^s, \text{pk}_\psi^v, m)$ 
     $\sigma_1 \leftarrow_{\$} \Psi'' \text{sim}(\text{sk}_\psi^v, \text{pk}_\psi^s, m)$ 
    Return  $\perp$ , if for any  $b' \in \{0, 1\}$ :  $\Psi'' \text{verify}(\text{pk}_\psi^v, \text{pk}_\psi^s, m, \sigma_{b'}) \neq 1$ 
    Return  $\sigma_b$ 
  return 1, if  $b^* = b$ 
  return 0

```

Fig. 12. Ψ'' Full Simulation-Based Non-Transferability

Definition 19 (Full Simulation-Based Non-Transferability). A designated verifier signature scheme Ψ'' satisfies full simulation-based non-transferability, if for all PPT adversaries $\mathcal{A}$ we have:

$$\left| \Pr \left[\mathbf{Exp}_{\mathcal{A}, \Psi''}^{\text{DVS-FSimNonTransfer}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda)$$

Construction. We now present our construction.

Security. We now investigate the security of our DVS Ψ'' in Construction 7.

Theorem 19. Let $\Psi'' = (\text{PG}_\psi'', \text{KGS}_\psi'', \text{KGV}_\psi'', \text{Sign}_\psi'', \text{Verify}_\psi'', \text{SIM}_\psi'')$ be the designated verifier signature scheme from Construction 7. If the NIZK proof system Π is simulation-sound extractable (SSE) and the function family $\mathcal{F} = \{f_\lambda\}$ is one-way, then Ψ'' is strongly unforgeable in the sense of Figure 2.

Intuition. Switch to an extraction-capable CRS (distance 0 if setup is already simulated). Run the extractor on any accepting forgery. If extraction fails, SSE breaks. If it succeeds, the witness w^* must satisfy $f(w^*) = y$, which inverts f .

The proof is given in Appendix L.1.

Theorem 20. Let $\Psi'' = (\text{PG}_\psi'', \text{KGS}_\psi'', \text{KGV}_\psi'', \text{Sign}_\psi'', \text{Verify}_\psi'', \text{SIM}_\psi'')$ be as defined in Construction 7. If the underlying Π satisfies trapdoor-detectable zero-knowledge, then Ψ'' satisfies full simulation-based non-transferability.

$\text{PG}_{\Psi}''(1^\lambda)$	: Return $\emptyset$.
$\text{KGS}_{\Psi}''(\text{pp}_{\Psi})$	: Sample $f \leftarrow_{\S} f_{\lambda}$, $w \leftarrow_{\S} C_{\lambda}$, and compute $y \leftarrow f(w)$. Return
	$(\text{sk}_{\Psi}^s, \text{pk}_{\Psi}^s) \leftarrow (w, (f, y))$
$\text{KGV}_{\Psi}''(\text{pp}_{\Psi})$	: Sample $(\text{crs}_{\Pi}; \xi) \leftarrow_{\S} \text{PG}_{\Pi}(1^\lambda, 0)$. Return
	$(\text{sk}_{\Psi}^v, \text{pk}_{\Psi}^v) \leftarrow (\xi, \text{crs}_{\Pi})$
$\text{Sign}_{\Psi}''(\text{sk}_{\Psi}^s, \text{pk}_{\Psi}^v, m)$	: Parse $\text{sk}_{\Psi}^s = w$, $\text{pk}_{\Psi}^v = \text{crs}_{\Pi}$, $\text{pk}_{\Psi}^s = (f, y)$. Compute
	$\sigma \leftarrow_{\S} \text{Prf}_{\Pi}(\text{crs}_{\Pi}, (f, y, m), w)$.
	Return σ .
$\text{Verify}_{\Psi}''(\text{pk}_{\Psi}^v, \text{pk}_{\Psi}^s, m, \sigma)$	: Parse $\text{pk}_{\Psi}^v = \text{crs}_{\Pi}$, $\text{pk}_{\Psi}^s = (f, y)$.
	Return
	$\text{Vfy}_{\Pi}(\text{crs}_{\Pi}, (f, y, m), \sigma)$
$\text{SIM}_{\Psi}''(\text{sk}_{\Psi}^v, \text{pk}_{\Psi}^s, m)$	: Parse $\text{sk}_{\Psi}^v = \xi$, $\text{pk}_{\Psi}^s = (f, y)$. Let $\text{crs}_{\Pi} \leftarrow_{\S} \text{PG}_{\Pi}(0; \xi)$ Check that
	$\text{DtctTd}_{\Pi}(\text{crs}_{\Pi}, \xi) = 0$. Derive $(\tau, \perp) \leftarrow_{\S} \text{SIM}_1(\xi)$. Return
	$\text{SIM}_2(\text{crs}_{\Pi}, \tau, (f, y, m))$

Construction 7: Designated Verifier Signature Scheme Ψ'' over Fixed L

Intuition. Only the LoR oracle depends on b . Use TDZK to replace honest proofs by simulated ones. With simulated proofs, the LoR answers become b -independent, so the advantage is 0.

The proof is given in Appendix L.2.

Acknowledgements. This work was in part supported by the European Union’s Horizon Europe project LICORICE (grant agreement no. 101168311) and the CHIST-ERA project REMINDER through the Austrian Science Fund (FWF) project number I 6650-N. Views and opinions expressed are however those of the authors only and do not necessarily reflect those of the funding agencies. Neither European Union nor the granting authorities can be held responsible for them.

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A Assumptions

We now present the assumptions we use in a formal way.

The Discrete Logarithm Assumption. We recall the standard discrete logarithm assumption in prime-order groups.

Let $\mathbb{G}$ be a cyclic group of prime order q , generated by an element $g \in \mathbb{G}$. The discrete logarithm assumption states that, for any PPT adversary $\mathcal{A}$, the probability of computing the exponent $x \in \mathbb{Z}_q$ from a given group element $h \leftarrow g^x$ is negligible:

Definition 20 (Discrete Logarithm Assumption). Let $(\mathbb{G}, q, g) \leftarrow_{\$} \text{GGen}(1^\lambda)$ be the output of a group generation algorithm that samples a cyclic group $\mathbb{G}$ of prime order q with generator g . Then, for all PPT adversaries $\mathcal{A}$, the advantage of solving the discrete logarithm problem is defined as:

$$\mathcal{A}_{\text{DL}}^{\mathbb{G}}(\lambda) := \Pr \left[(\mathbb{G}, q, g) \leftarrow_{\$} \text{GGen}(1^\lambda); x \leftarrow_{\$} \mathbb{Z}_q; h \leftarrow g^x \right. \\ \left. x^* \leftarrow_{\$} \mathcal{A}((\mathbb{G}, q, g, g^x)) : x^* = x \right] \leq \nu(\lambda)$$

B Additional Definitions

One-Way Functions. One-way functions formalize the intuitive notion of a function f that is easy to compute but hard to invert.

Let $\mathcal{F} = \{f_\lambda : C_\lambda \rightarrow D_\lambda\}_{\lambda \in \mathbb{N}}$ be a family of efficiently computable functions, where C_λ and D_λ are some domains (e.g., $\mathbb{Z}_q$ or $\{0, 1\}^n$). C_λ must be efficiently samplable.

Definition 21 (One-Way Function). (Intuition: hard to invert.) *We say that $\mathcal{F}$ is a one-way function family if for all PPT adversaries $\mathcal{A}$, the probability*

$$\Pr \left[x \leftarrow_{\$} C_\lambda; y \leftarrow f_\lambda(x); x' \leftarrow_{\$} \mathcal{A}(1^\lambda, y) : f_\lambda(x') = y \right] \leq \nu(\lambda)$$

is negligible.

Bilinear Pairings. Bilinear pairings are algebraic maps defined over groups equipped with an efficiently computable bilinear form. Concretely, let $\mathbb{G}_1$ be a cyclic group of prime order q , and let $\mathbb{G}_T$ be a cyclic group of the same order. A bilinear pairing is a function $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$ with the following properties:

- *Bilinearity:* For all $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_1$, and all $a, b \in \mathbb{Z}_q$, it holds that

$$e(P^a, Q^b) = e(P, Q)^{ab}.$$

- *Non-degeneracy:* If $g_1 \in \mathbb{G}_1$ and $g_t \in \mathbb{G}_T$ are generators, then $e(g_1, g_1)$ is a generator of $\mathbb{G}_T$.
- *Efficient computability:* The function e can be evaluated in time polynomial in $\log(q)$.

We use $(\mathbb{G}_1, \mathbb{G}_T, e, q, g_1, g_T) \leftarrow_{\$} \text{BGGen}(1^\lambda)$ to denote the group generation.

B.1 Pseudo-Random Functions

PRFs allow the holder of a, potentially secret, key κ to evaluate a function on an additional input point p .

Definition 22 (Pseudo-Random Functions). *A standard pseudo-random function PRF consists of two algorithms $\{\text{PRFKG}, \text{PRFEval}\}$ such that:*

PRFKG. *The algorithm PRFKG outputs a function key $\kappa \in \{0, 1\}^\lambda$, where λ is the security parameter:*

$$\kappa \leftarrow_{\$} \text{PRFKG}(1^\lambda)$$

PRFEval. *The deterministic algorithm PRFEval gets as input the secret key κ and a point $p \in \{0, 1\}^\lambda$ to evaluate. It outputs a new point $p' \in \{0, 1\}^\lambda$:*

$$p' \leftarrow \text{PRFEval}(\kappa, p)$$

```

Exp $\mathcal{A}, \text{PRF}$ PseudoRandomness( $\lambda$ )
 $\kappa \leftarrow_{\$} \text{PRFKG}(1^\lambda)$ 
 $b \leftarrow_{\$} \{0, 1\}$ 
 $f \leftarrow_{\$} F_\lambda$ 
 $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{PRFEval}}^\kappa}(1^\lambda)$ 
  where  $\mathcal{O}_{\text{PRFEval}}^\kappa(p)$ :
    return  $\perp$ , if  $p \notin \{0, 1\}^\lambda$ 
    if  $b = 0$ , return  $\text{PRFEval}(\kappa, p)$ 
    return  $f(p)$ 
  return 1, if  $b^* = b$ 
  return 0

```

Fig. 13. PRF Pseudo-Randomness

For security, it is required that PRF is actually pseudo-random.

Definition 23 (Pseudo-Randomness). A pseudo-random function PRF is pseudo-random, if for any PPT adversary $\mathcal{A}$ there exists a negligible function ν such that:

$$\left| \Pr[\mathbf{Exp}_{\mathcal{A}, \text{PRF}}^{\text{PseudoRandomness}}(\lambda) = 1] - \frac{1}{2} \right| \leq \nu(\lambda)$$

The corresponding experiment is depicted in Figure 13, where $F_\lambda = \{f : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda\}$ is the set of all functions f mapping a value $v \in \{0, 1\}^\lambda$ to another value $v' \in \{0, 1\}^\lambda$.

We sometimes map to group-elements instead of bitstrings.

B.2 Digital Signature Schemes

A digital signature scheme allows a party to sign messages such that the authenticity of the signature can be publicly verified.

Framework. We now present the formal framework for digital signature schemes.

Definition 24 (Digital Signature Scheme). A digital signature scheme Σ is a tuple of three PPT algorithms $(\text{KG}_\Sigma, \text{Sign}_\Sigma, \text{Verify}_\Sigma)$, such that:

KG_Σ . The algorithm KG_Σ takes as input a security parameter λ and outputs a signing key sk_Σ and a verification key pk_Σ :

$$(\text{sk}_\Sigma, \text{pk}_\Sigma) \leftarrow_{\$} \text{KG}_\Sigma(1^\lambda).$$

Sign_Σ . The signing algorithm Sign_Σ takes as input the secret key sk_Σ and a message m , and outputs a signature σ :

$$\sigma \leftarrow_{\$} \text{Sign}_\Sigma(\text{sk}_\Sigma, m).$$

Verify_Σ . The deterministic verification algorithm Verify_Σ takes as input the public key pk_Σ , a message m , and a signature σ , and outputs a bit $d \in \{0, 1\}$ indicating whether the signature is valid:

$$d \leftarrow \text{Verify}_\Sigma(\text{pk}_\Sigma, m, \sigma).$$

Definition 25 (Correctness). A digital signature scheme is called correct, if for all security parameters $\lambda \in \mathbb{N}$, for all $(\text{sk}_\Sigma, \text{pk}_\Sigma) \leftarrow_{\$} \text{KG}_\Sigma(1^\lambda)$, for all $m \in \{0, 1\}^*$, we have that

$$\text{Verify}_\Sigma(\text{pk}_\Sigma, m, \text{Sign}_\Sigma(\text{sk}_\Sigma, m)) = 1.$$

Security. We now define the security properties. Strong unforgeability for Σ requires that an adversary cannot come up with a message/signature pair on its own, even if it has adaptive oracle access to a signing oracle with messages of its own choice. In this paper, we only consider strong unforgeability.

```

Exp $\mathcal{A}, \Sigma$ StrongEUF-CMA( $\lambda$ )
   $(\text{sk}_\Sigma, \text{pk}_\Sigma) \leftarrow_{\$} \text{KG}_\Sigma(1^\lambda)$ 
   $\mathcal{Q} \leftarrow \emptyset$ 
   $(m^*, \sigma^*) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{Sign}}^{\text{sk}_\Sigma}}(\text{pk}_\Sigma)$ 
  where  $\mathcal{O}_{\text{Sign}}^{\text{sk}_\Sigma}(m)$ :
    let  $\sigma \leftarrow_{\$} \text{Sign}_\Sigma(\text{sk}_\Sigma, m)$ 
     $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$ 
    return  $\sigma$ 
  return 1, if  $\text{Verify}_\Sigma(\text{pk}_\Sigma, m^*, \sigma^*) = 1 \wedge (m^*, \sigma^*) \notin \mathcal{Q}$ 
  return 0

```

Fig. 14. Σ Strong Existential Unforgeability under Chosen Message Attacks

Definition 26 (Strong Unforgeability under Chosen Message Attacks). A digital signature scheme $\Sigma = (\text{KG}_\Sigma, \text{Sign}_\Sigma, \text{Verify}_\Sigma)$ is said to be strongly existentially unforgeable under chosen message attacks (Strong-EUF-CMA), if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:

$$\Pr \left[\mathbf{Exp}_{\mathcal{A}, \Sigma}^{\text{StrongEUF-CMA}}(\lambda) = 1 \right] \leq \nu(\lambda).$$

The corresponding experiment in Figure 14.

B.3 Non-Interactive Commitment Schemes with Checkable Public Parameters

Non-interactive commitment schemes with checkable public parameters enable a party to bind itself to a chosen value while keeping it hidden from the receiver.

At a later point, the committer can reveal auxiliary information - commonly referred to as the opening O - that allows the receiver to verify the correctness of the committed value [2].

For our case, we also require that the public parameters of the commitment scheme can be verified for well-formedness. This is formalized by the interface CP_Γ .

Framework. We now present the framework.

Definition 27 (Non-Interactive Commitment Schemes with Checkable Public Parameters). A non-interactive commitment scheme Γ for language L consists of four algorithms $\{\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma\}$, such that:

PG_Γ . The algorithm PG_Γ outputs public parameters of the scheme, where λ is the security parameter:

$$\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda).$$

CP_Γ . The algorithm CP_Γ outputs a bit $d \in \{0, 1\}$, verifying the validity of the public parameters pp_Γ :

$$d \leftarrow_{\$} \text{CP}_\Gamma(\text{pp}_\Gamma).$$

Cmt_Γ . The algorithm Cmt_Γ outputs a commitment C , along with opening information O , on input of a message m :

$$(C, O) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m).$$

Opn_Γ . The algorithm Opn_Γ verifies the validity of an opening O w.r.t. C and m , outputting a bit $d \in \{0, 1\}$:

$$d \leftarrow \text{Opn}_\Gamma(\text{pp}_\Gamma, C, O, m).$$

Definition 28 (Correctness). A non-interactive commitment scheme $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ is called correct if, for all $\lambda \in \mathbb{N}$, for all $\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda)$, for all $m \in \mathcal{M}$, for all $(C, O) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$, it holds that

$$\text{Opn}_\Gamma(\text{pp}_\Gamma, C, O, m) = 1.$$

and

$$\text{CP}_\Gamma(\text{pp}_\Gamma) = 1.$$

Security. We now define what security properties we require from Γ . Those include binding, hiding, and hiding against adverserially chosen public parameters.

Binding. Binding requires that an adversary cannot find two openings O_0 and O_1 such that a commitment C opens to two messages m_0 and m_1 such that $m_0 \neq m_1$.

Definition 29 (Binding). A non-interactive commitment scheme $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ is said to be computationally binding if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:

$$\Pr \left[\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{Bind}}(\lambda) = 1 \right] \leq \nu(\lambda),$$

where the experiment $\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{Bind}}(\lambda)$ is defined in Figure 15:

```

Exp $\mathcal{A}, \Gamma$ Bind( $\lambda$ )
   $\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda)$ 
   $(C, O_0, m_0, O_1, m_1) \leftarrow_{\$} \mathcal{A}(\text{pp}_\Gamma)$ 
  return 1, if  $m_0 \neq m_1 \wedge \text{Opn}_\Gamma(\text{pp}_\Gamma, C, O_0, m_0) = 1 \wedge \text{Opn}_\Gamma(\text{pp}_\Gamma, C, O_1, m_1) = 1$ 
  return 0

```

Fig. 15. Γ Binding

Computational Hiding. Computational Hiding requires that a commitment C does not reveal which message it contains, even if the adversary can choose the message. We sometimes shorten Computational Hiding to “Hiding”.

Definition 30 (Hiding). *A non-interactive commitment scheme $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ is said to be (computationally) hiding if for every PPT adversary $\mathcal{A}$, there exists a negligible function ν such that:*

$$\left| \Pr \left[\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{Hide}}(\lambda) = 1 \right] - \frac{1}{2} \right| \leq \nu(\lambda),$$

where the experiment $\mathbf{Exp}_{\mathcal{A}, \Gamma}^{\text{Hide}}(\lambda)$ is given in Figure 16:

```

Exp $\mathcal{A}, \Gamma$ Hide( $\lambda$ )
   $\text{pp}_\Gamma \leftarrow_{\$} \text{PG}_\Gamma(1^\lambda)$ 
   $b \leftarrow_{\$} \{0, 1\}$ 
   $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{Cmt}_\Gamma}^b}(\text{pp}_\Gamma)$ 
  where oracle  $\mathcal{O}_{\text{Cmt}_\Gamma}^b(m_0, m_1)$ :
     $(C_0, O_0) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m_0)$ 
     $(C_1, O_1) \leftarrow_{\$} \text{Cmt}_\Gamma(\text{pp}_\Gamma, m_1)$ 
    return  $\perp$  if  $C_0 = \perp$  or  $C_1 = \perp$ 
    return  $C_b$ 
  return 1 if  $b^* = b$ 
  return 0

```

Fig. 16. Γ Hiding experiment for non-interactive commitment schemes (honest pp_Γ)

B.4 Non-Interactive Proof Systems

Let L be an NP-language with associated witness relation R . A non-interactive proof system allows to prove membership of a statement x in L without interaction between prover and verifier.

Parameterized NP-Languages. In many applications, the relation underlying a NIZK proof system depends on externally chosen parameters (e.g., a cryptographic function f or a commitment scheme Γ). In such cases, the language is modeled as a family of NP-languages $L^{\mathcal{F}}$, indexed by a parameter, e.g., $f \in \mathcal{F}$. Formally:

$$L^{\mathcal{F}} := \{((f, x), w) \mid R_f(x, w) = 1\},$$

where each R_f is a relation depending on f , and the pair (f, x) is considered the statement.

The parameter f is typically chosen during key generation (or implicitly via the CRS), and hence is considered fixed for the duration of the proof system's usage. All notions of security for NIZKs, are understood to hold relative to the induced (parameterized) language $L^{\mathcal{F}}$.

We omit this parameterization in notation when unambiguous or not used; In more detail, we also already assume that the constructions itself are parametrized by the language L , i.e., we actually talk about families of constructions.

Framework. We now define the syntax for non-interactive proof systems.

Definition 31 (Non-Interactive Proof System). *A non-interactive proof system Π for language L consists of three PPT algorithms $\{\text{PG}_{\Pi}, \text{Prf}_{\Pi}, \text{Vfy}_{\Pi}\}$, such that:*

PG_{Π} . *The algorithm PG_{Π} outputs public parameters of the scheme, where λ is the security parameter:*

$$\text{crs}_{\Pi} \leftarrow_{\$} \text{PG}_{\Pi}(1^{\lambda}).$$

Prf_{Π} . *The algorithm Prf_{Π} outputs the proof π , on input of the CRS crs_{Π} , statement x to be proven, and the corresponding witness w :*

$$\pi \leftarrow_{\$} \text{Prf}_{\Pi}(\text{crs}_{\Pi}, x, w).$$

Vfy_{Π} . *The deterministic algorithm Vfy_{Π} verifies the proof π by outputting a bit $d \in \{0, 1\}$, w.r.t. to some CRS crs_{Π} and some statement x :*

$$d \leftarrow \text{Vfy}_{\Pi}(\text{crs}_{\Pi}, x, \pi).$$

Definition 32 (Correctness). *A non-interactive proof system is called correct, if for all $\lambda \in \mathbb{N}$, for all $\text{crs}_{\Pi} \leftarrow_{\$} \text{PG}_{\Pi}(1^{\lambda})$, for all $x \in L$, for all w such that $R(x, w) = 1$, for all $\pi \leftarrow_{\$} \text{Prf}_{\Pi}(\text{crs}_{\Pi}, x, w)$, it holds that $\text{Vfy}_{\Pi}(\text{crs}_{\Pi}, x, \pi) = 1$.*

In the context of (zero-knowledge) proof-systems, correctness is sometimes also referred to as completeness. In addition, we require two standard security notions for zero-knowledge proofs of knowledge: zero-knowledge and simulation-sound extractability. We define them analogously to the definitions given in [9].

Security. We now present the security guarantees we require.

Zero-Knowledge. Informally speaking, zero-knowledge says that the receiver of the proof π does not learn anything except the validity of the statement.

```

Exp $\mathcal{A}, \Pi, \text{SIM}$ Zero-Knowledge( $\lambda$ )
  ( $\text{crs}_\Pi, \tau$ )  $\leftarrow_{\$}$   $\text{SIM}_1(1^\lambda)$ 
   $b \leftarrow_{\$} \{0, 1\}$ 
   $b^* \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{P_b}(\cdot, \cdot)}(\text{crs}_\Pi)$ 
  where oracle  $\mathcal{O}_{P_0}(x, w)$ :
    return  $\perp$  if  $R(x, w) \neq 1$ 
    return  $\pi \leftarrow_{\$} \text{Prf}_\Pi(\text{crs}_\Pi, x, w)$ 
  and oracle  $\mathcal{O}_{P_1}(x, w)$ :
    return  $\perp$  if  $R(x, w) \neq 1$ 
    return  $\pi \leftarrow_{\$} \text{SIM}_2(\text{crs}_\Pi, \tau, x)$ 
  return 1, if  $b^* = b$ 
  return 0

```

Fig. 17. Π Zero-Knowledge

Definition 33 (Zero-Knowledge). A non-interactive proof system Π for language L is zero-knowledge, if for any PPT adversary $\mathcal{A}$, there exists an PPT simulator $\text{SIM} = (\text{SIM}_1, \text{SIM}_2)$ such that there exist a negligible function ν such that

$$\left| \Pr [\text{crs}_\Pi \leftarrow_{\$} \text{PG}_\Pi(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] - \Pr [(\text{crs}_\Pi, \tau) \leftarrow_{\$} \text{SIM}_1(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] \right| = 0,$$

and that

$$\left| \Pr [\text{Exp}_{\mathcal{A}, \Pi, \text{SIM}}^{\text{Zero-Knowledge}}(\lambda) = 1] - 1/2 \right| \leq \nu(\lambda),$$

where the corresponding experiment is depicted in Figure 17.

Simulation-sound Extractability. Simulation-sound extractability says that every adversary who is able to come up with a proof π^* for a statement must know the witness, even when seeing simulated proofs for adaptively chosen statements potentially not in L . Clearly, this implies that the proofs output by a simulation-sound extractable proof-systems are non-malleable. Note that the definition of simulation-sound extractability of [14] is stronger than the one used here.

Definition 34 (Simulation-Sound Extractability). A zero-knowledge non-interactive proof system Π for language L is said to be simulation-sound extractable, if for any PPT adversary $\mathcal{A}$, there exists a PPT extractor $\mathcal{E} = (\mathcal{E}_1, \mathcal{E}_2)$,


```

Exp $\mathcal{A}, \Pi, \mathcal{E}$ SimSoundExt( $\lambda$ )
  ( $\text{crs}_\Pi, \tau, \zeta$ )  $\leftarrow_{\$}$   $\mathcal{E}_1(1^\lambda)$ 
   $\mathcal{Q} \leftarrow \emptyset$ 
  ( $x^*, \pi^*$ )  $\leftarrow_{\$}$   $\mathcal{A}^{\mathcal{O}_{\text{SIM}}^\tau(\cdot)}(\text{crs}_\Pi)$ 
  where oracle  $\mathcal{O}_{\text{SIM}}^\tau(x)$ :
    obtain  $\pi \leftarrow_{\$}$   $\text{SIM}_2(\text{crs}_\Pi, \tau, x)$ 
     $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(x, \pi)\}$ 
  return  $\pi$ 
   $w^* \leftarrow_{\$}$   $\mathcal{E}_2(\text{crs}_\Pi, \zeta, x^*, \pi^*)$ 
  return 1, if  $\forall \text{f}_\Pi(x^*, \pi^*) = 1 \wedge R(x^*, w^*) = 0 \wedge (x^*, \pi^*) \notin \mathcal{Q}$ 
  return 0

```

Fig. 18. Π Simulation Sound Extractability

such that

$$\left| \Pr [(\text{crs}_\Pi, \tau) \leftarrow_{\$} \text{SIM}_1(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] - \Pr [(\text{crs}_\Pi, \tau, \zeta) \leftarrow_{\$} \mathcal{E}_1(1^\lambda) : \mathcal{A}(\text{crs}_\Pi) = 1] \right| = 0,$$

and that there exist a negligible function ν so that

$$\Pr [\mathbf{Exp}_{\mathcal{A}, \Pi, \mathcal{E}}^{\text{SimSoundExt}}(\lambda)] = 1 \leq \nu(\lambda),$$

where $\text{SIM} = (\text{SIM}_1, \text{SIM}_2)$ is as in Definition 33 and the corresponding experiment is depicted in Figure 18.

C Construction of Commitment Scheme

In particular, in a normal setting, the Pedersen parameters are chosen such that g and $h = g^x$ are generators of order q , while x is unknown to the committing party. This is clearly not possible, if both values are chosen by the adversary. In the extreme case that h only creates the trivial subgroup $1_{\mathbb{G}}$, the hiding property vanishes completely.

To tackle this “trivial” attack, one has to check whether the public parameters do have the desired structure, meaning that each generator is of a particular order.

Security of Construction 8.

Theorem 21. *Let $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ be the scheme from Construction 8. Then Γ is perfectly correct.*

$\text{PG}_\Gamma(1^\lambda)$: Generate a cyclic group $\mathbb{G}$ of prime order q with generator $g \in \mathbb{G}$, where the discrete logarithm problem is assumed to be hard. Sample $x \leftarrow_{\$} \mathbb{Z}_q$ and compute $h \leftarrow g^x$. Return the public parameters $\text{pp}_\Gamma \leftarrow (\mathbb{G}, q, g, h)$.
 $\text{CP}_\Gamma(\text{pp}_\Gamma)$: Parse pp_Γ as $(\mathbb{G}, q, g, h)$. Check that:

- $\mathbb{G}$ is a cyclic group of prime order q ,
- $g \in \mathbb{G}, h \in \mathbb{G}$,
- $g^q = 1, h^q = 1$,
- $g \neq 1, h \neq 1$.

 Return 1 if all checks pass; otherwise, return 0.
 $\text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$: Parse pp_Γ as $(\mathbb{G}, q, g, h)$.

- Return $\perp$ if $\text{CP}_\Gamma(\text{pp}_\Gamma) \neq 1$.
- Return $\perp$ if $m \notin \mathbb{Z}_q$. Note, the domain of messages can easily be expanded using a collision-resistant hash-function.
- Sample $r \leftarrow_{\$} \mathbb{Z}_q$.

 Compute the commitment as:

$$C \leftarrow g^m h^r \in \mathbb{G}.$$
 Return (C, r) .
 $\text{Opn}_\Gamma(\text{pp}_\Gamma, C, m, r)$: Parse pp_Γ as $(\mathbb{G}, q, g, h)$.

- Return 0 if $\text{CP}_\Gamma(\text{pp}_\Gamma) \neq 1$.
- Return 0 if $m \notin \mathbb{Z}_q$.
- Return 0 if $r \notin \mathbb{Z}_q$.

 Return 1 if

$$C = g^m h^r,$$
 and 0 otherwise.

Construction 8: Pedersen-Based Commitment Scheme Secure Against Adversarial Public Parameters

Proof. Follows by construction. $\square$

Theorem 22. *Let $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ be the commitment scheme from Construction 8. If the discrete logarithm (DL) problem is hard in the underlying group $\mathbb{G}$, then Γ is computationally binding.*

The proof is given in Appendix D.1.

Theorem 23. *Let $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ be the scheme from Construction 8, and assume the public parameters pp_Γ are honestly generated. Then Γ is perfectly hiding.*

The proof is given in Appendix D.2.

Theorem 24. *Let $\Gamma = (\text{PG}_\Gamma, \text{CP}_\Gamma, \text{Cmt}_\Gamma, \text{Opn}_\Gamma)$ be the scheme from Construction 8. Then Γ is perfectly hiding, even against adversarially chosen public parameters, as long as $\text{CP}_\Gamma(\text{pp}_\Gamma) = 1$.*

The proof is given in Appendix D.3.

D Proofs of the Commitment Scheme

D.1 Proof of Theorem 22 (Binding)

Hop Overview.

- **Game 0** Original binding experiment.
- **Game 1** Explicit parameter check.
- **Game 2** Embed DL challenge into pp_Γ .
- **Game 3** Extract DL from a double opening.

Proof. Let S_i denote adversarial success in Game i .

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Gamma}^{\text{Bind}}(\lambda)$. S_0 : $\mathcal{A}$ outputs two accepting openings (C, r_0, m_0) and (C, r_1, m_1) with $m_0 \neq m_1$.

Game 1: Explicit parameter check. As in Game 0, but abort if $\text{CP}_\Gamma(\text{pp}_\Gamma) \neq 1$.

Transition. Since PG_Γ outputs valid parameters,

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Game 2: Embedding the DL challenge. Replace PG_Γ by:

$\text{PG}'_\Gamma(1^\lambda) : \text{Sample } (\mathbb{G}, q, g, h = g^x) \leftarrow_{\S} \mathcal{C}^{\text{DL}} \text{ and set } \text{pp}_\Gamma \leftarrow (\mathbb{G}, q, g, h).$

Transition. This matches the distribution of PG_Γ (pick $x \leftarrow_{\S} \mathbb{Z}_q$, set $h = g^x$), hence

$$|\Pr[S_2] - \Pr[S_1]| = 0.$$

Game 3: Extract DL from a double opening. If $\mathcal{A}$ outputs (C, r_0, m_0, r_1, m_1) with $m_0 \neq m_1$ and both openings valid, then

$$C = g^{m_0} h^{r_0} = g^{m_1} h^{r_1} \Rightarrow g^{m_0 + xr_0} = g^{m_1 + xr_1} \Rightarrow m_0 + xr_0 \equiv m_1 + xr_1 \pmod{q}.$$

Thus $(r_1 - r_0) \not\equiv 0 \pmod{q}$ (else $m_0 \equiv m_1$), so

$$x \equiv (m_0 - m_1) \cdot (r_1 - r_0)^{-1} \pmod{q}.$$

Let S_3 be this event. Then $\Pr[S_3] = \Pr[S_2]$, and $\mathcal{B}$ solves DL whenever S_3 occurs.

Advantage Calculation. If $\Pr[S_0] = \nu(\lambda)$, identical-distribution hops yield

$$\Pr[\mathcal{B}^{\mathcal{A}} \text{ solves DL}] \geq \nu(\lambda).$$

By DL hardness, $\nu(\lambda) \leq \nu_{\text{DL}}(\lambda)$, hence

$$\Pr[\text{Exp}_{\mathcal{A}, \Gamma}^{\text{Bind}}(\lambda) = 1] \leq \nu_{\text{DL}}(\lambda).$$

□

Conclusion. Any successful double opening yields the discrete logarithm x . Thus the scheme is binding under DL.

D.2 Proof of Theorem 23 (Hiding)

Hop Overview.

- **Game 0** (Original hiding experiment).
- **Game 1** (Replace the commitment by a group-uniform element independent of b).

Proof. Proof Environment. Let $\mathcal{A}$ be PPT. Let S_i denote success in Game i (i.e., $b^* = b$). Work in a cyclic group $\mathbb{G} = \langle g \rangle$ of prime order q . Public key $h = g^x$ with fixed $x \in \mathbb{Z}_q$ and $x \neq 0$.

Game 0: Original experiment. The oracle $\mathcal{O}_{\text{Cmt}}^b(m_0, m_1)$ returns

$$C = g^{m_b} h^r = g^{m_b + xr}, \quad r \leftarrow_{\$} \mathbb{Z}_q.$$

Let $S_0 : b^* = b$.

Game 1: Uniform commitment independent of b . Same interface, but return a b -independent commitment:

$\mathcal{O}_{\text{Cmt}}^b(m_0, m_1)$: – Sample $r \leftarrow_{\$} \mathbb{Z}_q$.
 – Compute $C \leftarrow h^r = g^{xr}$.
 – Return C .

Let $S_1 : b^* = b$.

Transition $0 \rightarrow 1$. Since $x \neq 0$ and $r \leftarrow_{\$} \mathbb{Z}_q$, the variable xr is uniform in $\mathbb{Z}_q$. Adding the fixed shift m_b preserves uniformity, and $e \mapsto g^e$ is a bijection on $\mathbb{G}$. Hence $g^{m_b + xr}$ and g^{xr} are identically distributed and independent of m_b :

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Advantage Calculation. In Game 1 the view of $\mathcal{A}$ is independent of b . Therefore

$$|\Pr[S_1] - \tfrac{1}{2}| = 0 \quad \Rightarrow \quad |\Pr[S_0] - \tfrac{1}{2}| = 0.$$

□

Conclusion. The commitment is perfectly hiding: the distribution of C is group-uniform and independent of b .

D.3 Proof of Theorem 24 (Hiding vs. Adversarial Parameters)

Hop Overview.

- **Game 0** (Original adversarial-parameter hiding experiment).
- **Game 1** (Replace the commitment by a group-uniform element independent of b ; enforce $\text{CP}_r(\text{pp}_r) = 1$).

Proof. Proof Environment. Let $\mathcal{A}$ be PPT. The adversary chooses pp_r for each interaction with $\mathcal{O}_{\text{Cmt}}^b$. Let S_i denote success in Game i (i.e., $b^* = b$). Work in a cyclic group $\mathbb{G} = \langle g \rangle$ of prime order q . If $\text{CP}_r(\text{pp}_r) = 1$, then pp_r fixes $h = g^x$ with $x \in \mathbb{Z}_q$ and $x \neq 0$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Gamma}^{\text{AdvHide}}(\lambda)$. For input $(\text{pp}_\Gamma, m_0, m_1)$ with $\text{CP}_\Gamma(\text{pp}_\Gamma) = 1$, the oracle returns

$$C = g^{m_b} h^r = g^{m_b + xr}, \quad r \leftarrow_{\$} \mathbb{Z}_q.$$

Let $S_0 : b^* = b$.

Game 1: Uniform commitment independent of b . Same interface; reject malformed parameters and return a b -independent value:

$\mathcal{O}_{\text{Cmt}_\Gamma}^b(\text{pp}_\Gamma, m_0, m_1)$:
 – If $\text{CP}_\Gamma(\text{pp}_\Gamma) \neq 1$, return $\perp$.
 – Sample $r \leftarrow_{\$} \mathbb{Z}_q$; compute $C \leftarrow h^r = g^{xr}$; return C .

Let $S_1 : b^* = b$.

Transition 0 $\rightarrow$ 1. Since $x \neq 0$ and $r \leftarrow_{\$} \mathbb{Z}_q$, the variable xr is uniform in $\mathbb{Z}_q$. Adding the fixed shift m_b preserves uniformity; $e \mapsto g^e$ is a bijection on $\mathbb{G}$. Thus $g^{m_b + xr}$ and g^{xr} are identically distributed and independent of m_b :

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Advantage Calculation. In Game 1 the oracle output is independent of b . Hence

$$|\Pr[S_1] - \frac{1}{2}| = 0 \quad \Rightarrow \quad |\Pr[S_0] - \frac{1}{2}| = 0.$$

□

Conclusion. Even with adversarial pp_Γ satisfying CP_Γ , the commitment is group-uniform and independent of b ; the scheme is perfectly hiding against AdvHide.

E Proof of Theorem 10 (Limit of CH)

Hop Overview.

- **Game 0** (original F-lnd experiment).
- **Game 1** (semantic rewrite with extraction; simulated proofs).
- **Game 1'** (CZK-hybrid: replace simulated by real proofs).
- **Game 2** (ideal-case analysis with extractor correctness and SSE implication).

Proof. Proof Environment. Let $\mathcal{A}$ be PPT and let S_i denote success in Game i . Let $(\text{crs}_\Pi, \tau, \zeta) \leftarrow_{\$} \mathcal{E}_1(1^\lambda)$ be the CRS; by SSE-setup the distribution equals the simulated/real setup, so no hop is needed. Maintain the simulator-proof set $\mathcal{Q}$ as in Fig. 18.

Game 0: Original experiment.

$$\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-lnd}}(\lambda), \quad S_0 : b^* = b.$$

Game 1: Semantic rewrite with extraction (simulated proofs). Same interface; describe $\mathcal{A}$'s effective strategy:

- Submit (m, m') ; receive (h, r) .
- Set $x \leftarrow (\text{pp}_r, h, m')$.
- Compute $O^* \leftarrow_{\S} \mathcal{E}_2(\text{crs}_{\Pi}, \zeta, x, r)$.
- Output $b^* \leftarrow 0$ iff $\text{Opn}_r(\text{pp}_r, h, m', O^*) = 1$, else 1.

Transition $0 \rightarrow 1$. Pure rewrite, so $|\Pr[S_1] - \Pr[S_0]| = 0$.

Game 1': CZK-hybrid (sim $\rightarrow$ real). Replace every simulated proof by a real prover proof.

Transition $1 \rightarrow 1'$. By CZK,

$$|\Pr[S_{1'}] - \Pr[S_1]| \leq \nu_{\text{czk}}(\lambda).$$

Game 2: Ideal-case analysis with extraction. Now all proofs are real.

- If $b = 0$: (h, r) honest via CHHash. By correctness of $\mathcal{E}_2$, extraction succeeds except with ν_{sse} , so $b^* = b$ w.p. $\geq 1 - \nu_{\text{sse}}$.
- If $b = 1$: an accepting fresh $(x^*, \pi^*) \notin \mathcal{Q}$ with failed extraction would contradict SSE.

SSE implication (for $1' \rightarrow 2$).

SSE implication. If there exists $(x^*, \pi^*) \notin \mathcal{Q}$ with $\text{Vfy}_{\Pi}(\text{crs}_{\Pi}, x^*, \pi^*) = 1$ and $\mathcal{E}_2(\text{crs}_{\Pi}, \zeta, x^*, \pi^*)$ fails, then an adversary wins $\text{Exp}_{\Pi, \mathcal{E}}^{\text{SimSoundExt}}(\lambda)$. Thus this bad event has probability at most $\nu_{\text{sse}}(\lambda)$.

Advantage Analysis. By total probability and the transitions,

$$\Pr[S_2] \geq 1 - \nu_{\text{czk}}(\lambda) - \nu_{\text{sse}}(\lambda) \Rightarrow \left| \Pr[S_0] - \frac{1}{2} \right| \geq \frac{1}{2} - \nu(\lambda).$$

□

Conclusion. CZK carries sim $\rightarrow$ real; SSE rules out “verify without extraction”. Both are negligible, so CH' violates Full-Indistinguishability.

F Proof of Theorem 1 (Strong Unforgeability)

Hop Overview.

- **Game 0** Original experiment.
- **Game 1** Extractable CRS: $(\text{crs}_{\Pi}, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$.
- **Game 2** Extraction on the adversary's forgery; “accept & extraction fails” breaks SSE.

Proof. Let S_i denote success in Game i . A forgery attempt (m^*, σ^*) is called *fresh* iff $(m^*, \sigma^*) \notin \mathcal{Q}_{\text{sign}}$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Sigma}^{\text{SUF-CMA}}(\lambda)$ with CRS $(\text{crs}_{\Pi}, \tau) \leftarrow \text{SIM}_1(1^\lambda)$. Signing uses $\text{SIM}_2(\text{crs}_{\Pi}, \tau, \cdot)$ and records $\mathcal{Q}_{\text{sign}}$. Let S_0 be the event that $\mathcal{A}$ outputs a fresh (m^*, σ^*) with $\text{Verify}_{\Sigma}(\text{pk}, m^*, \sigma^*) = 1$. Parse σ^* as π^* proving statement x^* (i.e., relation and m^*).

Game 1: Extractable CRS. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$. All else remains unchanged. Let S_1 be success here.

Transition $0 \rightarrow 1$. Identical distribution on (crs_Π, τ) yields $|\Pr[S_1] - \Pr[S_0]| = 0$.

Game 2: Extraction on forgery. On a candidate forgery, run

$$w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x^*, \pi^*).$$

Let S_2 be the event either extraction fails or a wrong statement was proven.

Transition $1 \rightarrow 2$. If $\text{Vfy}_\Pi(\text{crs}_\Pi, x^*, \pi^*) = 1$ and $(x^*, \pi^*) \notin \mathcal{Q}$ but $\mathcal{E}_2$ fails, then $\mathcal{B}_{\text{sse}}$ wins simulation-sound extractability. Hence

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{sse}}(\lambda).$$

$\mathcal{B}_{\text{sse}}(1^\lambda)$

1. Receive crs_Π and oracle $\mathcal{O}_{\text{SIM}}^\tau(\cdot)$ from the SSE challenger; maintain the set $\mathcal{Q}$ of simulator-issued pairs.
2. Run $\mathcal{A}$; whenever a proof for statement x is needed by signing, query $\mathcal{O}_{\text{SIM}}^\tau(x)$, get π , return it, and record $(x, \pi) \in \mathcal{Q}$.
3. Upon a valid fresh forgery with NIZK part π^* for x^* where $(x^*, \pi^*) \notin \mathcal{Q}$, if extraction fails while verify accepts, output (x^*, π^*) to the challenger.

Advantage analysis. In our construction, honest signatures use simulated proofs, i.e., no real witness is embedded; any accepting fresh proof without an extractable witness triggers the SSE bad event, so $\Pr[S_2] \leq \nu_{\text{sse}}(\lambda)$. Therefore

$$\Pr[S_0] \leq \nu_{\text{sse}}(\lambda).$$

□

Conclusion. Either there exists an accepting fresh proof without an extractable witness (violating SSE), or forging is infeasible. Hence Σ is strongly unforgeable.

G Proofs of first DVS

G.1 Proof of Theorem 3 (DVS Strong Unforgeability)

Hop Overview.

- **Game 0** Original experiment.
- **Game 1** Extractable CRS: $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$.
- **Game 2** Run extractor on the forgery; if Verify accepts but extraction fails or a wrong statement was proven, SSE is broken.
- **Game 3** If the extracted w^* satisfies $f(w^*) = y$, invert f .

Proof. Let S_i denote success in Game i . A forgery attempt (m^*, σ^*) is *fresh* iff $(m^*, \sigma^*) \notin \mathcal{Q}_{\text{sign}}$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Psi}^{\text{SUF-CMA}}(\lambda)$ with CRS $(\text{crs}_\Pi, \tau) \leftarrow \text{SIM}_1(1^\lambda)$. Signing uses the scheme and records $\mathcal{Q}_{\text{sign}}$. Let S_0 be the event that $\mathcal{A}$ outputs a fresh (m^*, σ^*) with $\text{Verify}_\Psi(\text{pk}_\Psi^v, \text{pk}_\Psi^s, m^*, \sigma^*) = 1$. Parse σ^* to its NIZK part π^* for statement x^* .

Game 1: Extractable CRS. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$. Rest unchanged. Let S_1 denote success here.

Transition $0 \rightarrow 1$. Identical distribution on (crs_Π, τ) yields $|\Pr[S_1] - \Pr[S_0]| = 0$.

Game 2: Extraction. On a candidate forgery, run

$$w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x^*, \pi^*).$$

Let S_2 be the event “ $\text{Vfy}_\Pi(\text{crs}_\Pi, x^*, \pi^*) = 1$ and either extraction fails or the extracted witness does not satisfy the relation.”

Transition $1 \rightarrow 2$. If $\text{Vfy}_\Pi(\text{crs}_\Pi, x^*, \pi^*) = 1$ on a fresh (x^*, π^*) yet extraction fails or is wrong, the SSE experiment is broken. Hence

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{sse}}(\lambda).$$

$\mathcal{B}_{\text{sse}}(1^\lambda)$ 1. Receive crs_Π *only* from the SSE challenger; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$.
 2. Choose $f \in \mathcal{F}$, sample $w \leftarrow C_\lambda$, set $y \leftarrow f(w)$ and publish $\text{pk}_\Psi^s \leftarrow (f, y)$.
 3. Answer signing queries on m by producing a *real* NIZK proof $\pi \leftarrow \text{Prf}_\Pi(\text{crs}_\Pi, (f, y, m), w)$ and return σ ; record $(m, \sigma) \in \mathcal{Q}_{\text{sign}}$.
 4. Run $\mathcal{A}$; on a fresh accepting forgery (m^*, σ^*) with NIZK part π^* for $x^* = (f, y, m^*)$, output (x^*, π^*) to the SSE challenger.

Game 3: Inversion check. Define S_3 as “extraction succeeds and $f(w^*) = y$ ”. Whenever S_3 occurs, we invert f .

$\mathcal{B}_{\text{owf}}^A(f, y)$ 1. Receive OWF challenge (f, y) .
 2. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$ and $\text{pk}_\Psi^s \leftarrow (f, y)$.
 3. Answer signing and simulation queries using $\text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m))$.
 4. Run $\mathcal{A}$; on forgery (m^*, σ^*) compute $w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x^*, \pi^*)$ with $x^* = (f, y, m^*)$. If $f(w^*) = y$, output w^* .

Advantage analysis. If $\text{Verify}_\Psi = 1$ yet extraction fails (or yields an invalid witness) on a fresh proof, $\mathcal{B}_{\text{sse}}$ succeeds; thus $|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{sse}}$. Whenever S_3 occurs, $\mathcal{B}_{\text{owf}}$ inverts f , so $\Pr[S_3] \leq \nu_{\text{owf}}$. Therefore

$$\Pr[S_0] \leq \nu_{\text{sse}}(\lambda) + \nu_{\text{owf}}(\lambda)$$

□

Conclusion. Either an accepting fresh proof lacks an extractable valid witness (SSE violation), or extraction yields w^* with $f(w^*) = y$ and inverts f . Hence Ψ is strongly unforgeable.

G.2 Proof of Theorem 4 (DVS Simulation-Based Non-Transferability)

Hop Overview.

- **Game 0** (Original experiment): $\text{Exp}^{\text{DVS-SimNonTransfer}}$ with real $\mathcal{O}_{\text{LoRSign}}$.
- **Game 1** (Simulated LoR oracle): replace every $\mathcal{O}_{\text{LoRSign}}$ answer by the simulator’s output; other oracles unchanged.

Proof. Let $\nu_i(\lambda)$ be the distinguishing advantage in Game i .

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Psi}^{\text{DVS-SimNonTransfer}}(\lambda)$ as in Fig. 3. Interfaces available: $\mathcal{O}_{\text{LoRSign}}^b$, $\mathcal{O}_{\text{Sign}\Psi}$, $\mathcal{O}_{\text{SIM}\Psi}$. Let $\nu_0(\lambda) = |\Pr[b^* = b] - \frac{1}{2}|$.

Game 1: Simulated LoR oracle. All answers of $\mathcal{O}_{\text{LoRSign}}$ are produced by SIM_2 on the same statements. Other oracles behave as in Game 0. Let $\nu_1(\lambda)$ be the advantage here.

Transition 0 $\rightarrow$ 1. If $\mathcal{A}$ distinguishes, $\mathcal{B}_{\text{zk}}$ breaks ZK by wiring $\mathcal{O}_{\text{LoRSign}}$ to the ZK left-or-right oracle:

$$|\nu_0(\lambda) - \nu_1(\lambda)| \leq \nu_{\text{zk}}(\lambda).$$

- | | |
|--------------------------------------|--|
| $\mathcal{B}_{\text{zk}}(1^\lambda)$ | <ol style="list-style-type: none"> 1. Receive crs_Π and access to the ZK challenger's $\mathcal{O}_{P_b}^r(\cdot)$; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$. 2. Choose $f \in \mathcal{F}$, sample $w \leftarrow C_\lambda$, set $y \leftarrow f(w)$; publish $(\text{pk}_\Psi^s, \text{pk}_\Psi^v) = ((f, y), \text{crs}_\Pi)$. 3. On $\mathcal{O}_{\text{LoRSign}}(m)$: query $\mathcal{O}_{P_b}^r((f, y, m), w)$ and return its output. 4. Answer $\mathcal{O}_{\text{Sign}\Psi}$, $\mathcal{O}_{\text{SIM}\Psi}$ honestly as specified. 5. Output b^* to the ZK challenger. |
|--------------------------------------|--|

Advantage Analysis. In Game 1, $\mathcal{O}_{\text{LoRSign}}$ outputs are simulator-generated and b -independent; other oracles are b -agnostic. Hence $\nu_1(\lambda) = 0$ and

$$\nu_0(\lambda) \leq \nu_{\text{zk}}(\lambda) = \nu(\lambda).$$

□

Conclusion. Distinguishing real vs. simulated $\mathcal{O}_{\text{LoRSign}}$ contradicts zero-knowledge of Π . Therefore Ψ satisfies simulation-based non-transferability.

H Proofs of First CH

H.1 Proof of Theorem 5 (Full Collision Resistance)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{F-CollRes}}$: adversary outputs $(m^*, r^*, m'^*, r'^*, h^*)$ with $m^* \neq m'^*$, both checks accept, and *at least one* of (h^*, m^*) , (h^*, m'^*) is not from $\mathcal{O}_{\text{CHAdapt}}$.
- **Game 1** Bookkeeping: classify accepting tuples as external (E) vs. simulated (S) and ensure $\text{E/S} \Rightarrow \text{E/E}$ using a recorded extra external tuple for the same h^* .
- **Game 2** Extractable CRS: $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$.
- **Game 3** Dual extraction on two external accepting tuples; any accept-but-fail implies an SSE break.
- **Game 4** Binding hand-off: two extracted openings for the same h^* .

Proof. Let S_i denote success in Game i . Maintain $\mathcal{Q} \subseteq \mathcal{H} \times \mathcal{M}$ for all (h, m) output by $\mathcal{O}_{\text{CHAdapt}}$. A tuple (h, m, r) is *external* iff $(h, m) \notin \mathcal{Q}$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-CollRes}}(\lambda)$ with $(\text{crs}_{\Pi}, \tau) \leftarrow \text{SIM}_1(1^\lambda)$, honest pp_{Γ} , and CHAdapt answered via $\text{SIM}_2(\text{crs}_{\Pi}, \tau, \cdot)$. Win if

$$\text{CHCheck}(\text{pk}_{\text{ch}}, m^*, r^*, h^*) = \text{CHCheck}(\text{pk}_{\text{ch}}, m'^*, r'^*, h^*) = 1, \quad m^* \neq m'^*,$$

and *at least one* of (h^*, m^*) , (h^*, m'^*) is external. Let S_0 be this event.

Game 1: Bookkeeping and E/S $\Rightarrow$ E/E. As in Game 0, but additionally record every *external* accepting tuple in

$$\mathcal{R} \leftarrow \mathcal{R} \cup \{(h, m, r) \mid \text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1 \wedge (h, m) \notin \mathcal{Q}\}.$$

If the final pair is of type E/S, replace the S-side by a distinct $(h^*, \tilde{m}, \tilde{r}) \in \mathcal{R}$ with $\tilde{m} \notin \{m^*, m'^*\}$ to obtain an E/E pair for h^* . Let S_1 be success (same winning predicate after replacement).

Transition 0 $\rightarrow$ 1. Pure logging and deterministic post-processing; distributions coincide:

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Game 2: Extractable CRS. Sample $(\text{crs}_{\Pi}, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; answer CHAdapt via $\text{SIM}_2(\text{crs}_{\Pi}, \tau, \cdot)$. Let S_2 be success.

Transition 1 $\rightarrow$ 2. Identical marginal on (crs_{Π}, τ) yields $|\Pr[S_2] - \Pr[S_1]| = 0$.

Game 3: Dual extraction on external tuples. With an E/E pair for h^* , set $x_1 = (\text{pp}_{\Gamma}, h^*, m^*)$, $x_2 = (\text{pp}_{\Gamma}, h^*, m'^*)$ and run

$$w_1 \leftarrow \mathcal{E}_2(\text{crs}_{\Pi}, \zeta, x_1, r^*), \quad w_2 \leftarrow \mathcal{E}_2(\text{crs}_{\Pi}, \zeta, x_2, r'^*).$$

Let S_3 be “both extractions succeed”.

Transition 2 $\rightarrow$ 3. If Verify accepts on an external $(x, \pi) \notin \mathcal{Q}$ but extraction fails, $\mathcal{B}_{\text{sse}}$ wins SimSoundExt . By a union bound over two external extractions,

$$|\Pr[S_3] - \Pr[S_2]| \leq 2\nu_{\text{sse}}^{(2)}(\lambda).$$

- $\mathcal{B}_{\text{sse}}^{(2)}(1^\lambda)$
1. Receive *only* crs_{Π} and the simulator oracle $\text{SIM}_2(\text{crs}_{\Pi}, \cdot)$; set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_{\Pi}, \text{pp}_{\Gamma})$. Maintain $\mathcal{Q}$ and $\mathcal{R}$.
 2. Answer CHAdapt by forwarding to the simulator and insert each returned (h, m) into $\mathcal{Q}$.
 3. Run $\mathcal{A}$; using Game 1’s rule, ensure two *external* accepting tuples (x_1, π_1) , $(x_2, \pi_2) \notin \mathcal{Q}$ for the same h^* .
 4. If extraction (performed by the challenger) fails on at least one external pair, choose uniformly one failing pair and return it. The $1/2$ selection gives the global factor 2 in the bound.

Game 4: Binding hand-off. On S_3 , output $(h^*, m^*, w_1, m'^*, w_2)$ to the binding challenger. Let S_4 denote success there.

Transition 3 $\rightarrow$ 4. Pure relabeling: $|\Pr[S_4] - \Pr[S_3]| = 0$.

Reduction to binding.

- $\mathcal{B}_{\text{bind}}(1^\lambda)$
1. Receive pp_{Γ} ; sample $(\text{crs}_{\Pi}, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_{\Pi}, \text{pp}_{\Gamma})$; answer CHAdapt via SIM_2 ; maintain $\mathcal{Q}, \mathcal{R}$.
 2. Run $\mathcal{A}$; apply the Game 1 E/S $\rightarrow$ E/E rule for h^* ; extract $w_1 \leftarrow \mathcal{E}_2(\text{crs}_{\Pi}, \zeta, (\text{pp}_{\Gamma}, h^*, m^*), r^*)$ and $w_2 \leftarrow \mathcal{E}_2(\text{crs}_{\Pi}, \zeta, (\text{pp}_{\Gamma}, h^*, m'^*), r'^*)$.
 3. Return $(h^*, m^*, w_1, m'^*, w_2)$ to the binding challenger.

Advantage analysis. Binding succeeds whenever S_4 occurs, hence

$$\Pr[S_0] \leq 2\nu_{\text{sse}}^{(2)}(\lambda) + \nu_{\text{bind}}(\lambda).$$

□

Conclusion. With bookkeeping that converts E/S to E/E, any successful collision either yields an accepting external proof without an extractable witness (SSE break) or two distinct openings for the same h^* (binding break).

H.2 Proof of Theorem 6 (Normal Indistinguishability)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{N-Ind}}$: adversary interacts with $\mathcal{O}_{\text{HashOrAdapt}}^b$ and guesses b .
- **Game 1** Replace all NIZK proofs produced by CHHash with simulated proofs via CZK.
- **Game 2** In all CHHash calls, fix the message to m' ; proofs remain simulated.

Proof. Let S_i be the event $\mathcal{A}$ outputs $b^* = b$ in Game i .

Game 0: Original experiment. Set $\nu_0(\lambda) = |\Pr[S_0] - \frac{1}{2}|$.

Game 1: Simulated proofs in CHHash (CZK). Inside $\mathcal{O}_{\text{HashOrAdapt}}^b$, answer $\text{CHHash}(m)$ by computing $C \leftarrow \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$ and $\pi \leftarrow \text{SIM}_2(\text{crs}_\Pi, \tau, x)$ for $x = (\text{pp}_\Gamma, C, m)$; return (C, π) . Let S_1 be success here.

Transition $0 \rightarrow 1$. By composable ZK,

$$|\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{czk}}(\lambda).$$

- $\mathcal{B}^{\text{CZK}}(1^\lambda)$

 1. Receive (crs_Π, τ) / CZK interface; set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$ for fresh pp_Γ .
 2. For each $\text{CHHash}(m)$ call: compute $C \leftarrow \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$, obtain π from the CZK simulator on $x = (\text{pp}_\Gamma, C, m)$, and return (C, π) .
 3. Forward other oracles as in Game 0; output b^* .

Game 2: Fix message in CHHash to m' (hiding). As in Game 1, but ignore b and always hash m' ; proofs remain simulated. Let S_2 be success.

Transition $1 \rightarrow 2$. By hiding of Γ ,

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{hide}}(\lambda).$$

- $\mathcal{B}^{\text{hide}}(1^\lambda)$

 1. Receive pp_Γ and access to the hiding challenger $\mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(\cdot, \cdot)$.
 2. Sample $(\text{crs}_\Pi, \tau) \leftarrow \text{SIM}_1(1^\lambda)$; set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$.
 3. On a challenge (m, m') from $\mathcal{A}$: query $C \leftarrow \mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(m', m')$ and simulate $\pi \leftarrow \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m'))$; return (C, π) .
 4. Relay other oracle answers as in Game 1; output b^* .

Advantage analysis. The view in Game 2 is independent of b , so $\Pr[S_2] = \frac{1}{2}$. Hence

$$|\Pr[S_0] - \frac{1}{2}| \leq \nu_{\text{czk}}(\lambda) + \nu_{\text{hide}}(\lambda) = \nu(\lambda).$$

□

Conclusion. After simulating all proofs in CHHash (CZK) and fixing the hashed message to m' (hiding), the view is b -independent. Any non-negligible advantage would violate CZK or commitment hiding.

I Proofs of Second CH

I.1 Proof of Theorem 8 (Full Collision Resistance)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{F-CollRes}}$: adversary outputs $(m^*, r^*, m'^*, r'^*, h^*)$ with $m^* \neq m'^*$, both checks accept, and *at least one* of (h^*, m^*) , (h^*, m'^*) is not from $\mathcal{O}_{\text{CHAdapt}}$.
- **Game 1** Bookkeeping: classify accepting tuples as external (E) vs. simulated (S) and deterministically turn E/S into E/E using a recorded extra external tuple for the same h^* .
- **Game 2** Extractable CRS: $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$.
- **Game 3** Dual extraction on two external accepting tuples; any accept-but-fail implies an SSE break.
- **Game 4** Binding hand-off: two extracted openings for the same h^* .

Proof. Let S_i denote success in Game i . Let $\mathcal{Q} \subseteq \mathcal{H} \times \mathcal{M}$ collect all (h, m) issued by $\mathcal{O}_{\text{CHAdapt}}$; a tuple (h, m, r) is *external* iff $(h, m) \notin \mathcal{Q}$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-CollRes}}(\lambda)$ with honest pp_Γ and $(\text{crs}_\Pi, \tau) \leftarrow \text{SIM}_1(1^\lambda)$. Win if $\text{CHCheck}(\text{pk}_{\text{ch}}, m^*, r^*, h^*) = \text{CHCheck}(\text{pk}_{\text{ch}}, m'^*, r'^*, h^*) = 1$, $m^* \neq m'^*$, and *at least one* of (h^*, m^*) , (h^*, m'^*) is external. Event S_0 .

Game 1: Bookkeeping and E/S $\Rightarrow$ E/E. As in Game 0, but record every external accepting tuple in

$$\mathcal{R} \leftarrow \mathcal{R} \cup \{(h, m, r) \mid \text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1 \wedge (h, m) \notin \mathcal{Q}\}.$$

If the final pair is E/S, replace the S-side by a distinct $(h^*, \tilde{m}, \tilde{r}) \in \mathcal{R}$ with $\tilde{m} \notin \{m^*, m'^*\}$, yielding an E/E pair for h^* . Let S_1 be success (same win predicate after replacement).

Transition 0 $\rightarrow$ 1. Pure logging and deterministic post-processing:

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Game 2: Extractable CRS. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$ with $(\text{crs}_\Pi, \tau) \equiv_d \text{SIM}_1(1^\lambda)$; answer CHAdapt via $\text{SIM}_2(\text{crs}_\Pi, \tau, \cdot)$. Event S_2 .

Transition 1 $\rightarrow$ 2. Identical marginal on (crs_Π, τ) :

$$|\Pr[S_2] - \Pr[S_1]| = 0.$$

Game 3: Dual extraction. With an E/E pair for h^* , set $x_1 = (\text{pp}_\Gamma, h^*, m^*)$, $x_2 = (\text{pp}_\Gamma, h^*, m'^*)$ and run

$$w_1 \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x_1, r^*), \quad w_2 \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x_2, r'^*).$$

Let S_3 be “both extractions succeed.”

Transition 2 $\rightarrow$ 3. If Verify accepts on an external $(x, \pi) \notin \mathcal{Q}$ but extraction fails, $\mathcal{B}_{\text{sse}}$ wins SimSoundExt . Union bound over two external extractions:

$$|\Pr[S_3] - \Pr[S_2]| \leq 2\nu_{\text{sse}}^{(2)}(\lambda).$$

- $\mathcal{B}_{\text{sse}}^{(2)}(1^\lambda)$
1. Receive *only* crs_Π and access to the simulator oracle $\text{SIM}_2(\text{crs}_\Pi, \cdot)$ from the SSE challenger. Sample honest pp_Γ and set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$. Maintain $\mathcal{Q}$ and $\mathcal{R}$.
 2. Answer CHAdapt by forwarding to SIM_2 and insert each returned (h, m) into $\mathcal{Q}$.
 3. Run $\mathcal{A}$; enforce the Game 1 E/S $\rightarrow$ E/E rule to obtain two *external* accepting pairs $(x_1, \pi_1), (x_2, \pi_2) \notin \mathcal{Q}$ for the same h^* .
 4. If extraction fails on at least one external pair, choose uniformly one failing pair (x_b, π_b) and return it. The 1/2 selection yields the global factor 2 in the bound.

Game 4: Binding hand-off. Return $(h^*, m^*, w_1, m'^*, w_2)$ to the commitment-binding challenger. Event S_4 .

Transition $3 \rightarrow 4$. The step $3 \rightarrow 4$ is a pure relabeling: $|\Pr[S_4] - \Pr[S_3]| = 0$.

Reduction to binding.

- $\mathcal{B}_{\text{bind}}(1^\lambda)$
1. Receive pp_Γ from the binding challenger. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$ and set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$.
 2. Answer CHAdapt via $\text{SIM}_2(\text{crs}_\Pi, \tau, \cdot)$; maintain $\mathcal{Q}$ and $\mathcal{R}$.
 3. Run $\mathcal{A}$; apply the Game 1 E/S $\rightarrow$ E/E rule; extract $w_1 \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, (\text{pp}_\Gamma, h^*, m^*), r^*)$ and $w_2 \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, (\text{pp}_\Gamma, h^*, m'^*), r'^*)$.
 4. Output $(h^*, m^*, w_1, m'^*, w_2)$ to the binding challenger.

Advantage analysis. Binding succeeds on S_4 , hence

$$\Pr[S_0] \leq 2\nu_{\text{sse}}^{(2)}(\lambda) + \nu_{\text{bind}}(\lambda).$$

□

Conclusion. With bookkeeping that reduces E/S to E/E, any successful collision either yields an accepting external proof without an extractable witness (SSE break) or two distinct openings for the same h^* (binding break).

I.2 Proof of Theorem 9 (Indistinguishability)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{S-Ind}}$: CHKG via SIM_1 ; CHAdapt via SIM_2 ; CHHash uses real proofs.
- **Game 1** Replace all proofs output by CHHash with simulated ones via SIM_2 .
- **Game 2** In $\mathcal{O}_{\text{HashOrAdapt}}$, always commit to m' ; proofs remain simulated.

Proof. Let S_i be the event $\mathcal{A}$ outputs $b^* = b$ in Game i .

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \text{CH}'}^{\text{S-Ind}}(\lambda)$ as in Fig. 7. Event S_0 .

Game 1: Simulated proofs in CHHash (CZK). For each $\text{CHHash}(m)$: compute $C \leftarrow \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$ and $\pi \leftarrow \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m))$; return (C, π) . Event S_1 .

Transition $0 \rightarrow 1$. By composable ZK,

$$|\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{czk}}(\lambda).$$

- $\mathcal{B}^{\text{CZK}}(1^\lambda)$
1. Receive (crs_Π, τ) . Set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$ for fresh pp_Γ .
 2. On each $\text{CHHash}(m)$: compute $C \leftarrow \text{Cmt}_\Gamma(\text{pp}_\Gamma, m)$, obtain π from the CZK simulator for $x = (\text{pp}_\Gamma, C, m)$, and return (C, π) .
 3. Relay other oracles as in Game 0; output b^* .

Game 2: Message fixing (hiding). In $\mathcal{O}_{\text{HashOrAdapt}}$, ignore b and commit to m' ; simulate proofs as above. Event S_2 .

Transition $1 \rightarrow 2$. By hiding of Γ ,

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{hide}}(\lambda).$$

$\mathcal{E}^{\text{hide}}(1^\lambda)$

1. Receive pp_Γ and access to the hiding challenger $\mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(\cdot, \cdot)$.
2. Sample $(\text{crs}_\Pi, \tau) \leftarrow \text{SIM}_1(1^\lambda)$; set $\text{pk}_{\text{ch}} \leftarrow (\text{crs}_\Pi, \text{pp}_\Gamma)$.
3. On a challenge (m, m') from $\mathcal{A}$: query $C \leftarrow \mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(m', m')$ and simulate $\pi \leftarrow \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m'))$; return (C, π) .
4. Relay other oracle answers as in Game 1; output b^* .

Advantage analysis. Game 2 is b -independent, so $\Pr[S_2] = \frac{1}{2}$. Hence

$$|\Pr[S_0] - \frac{1}{2}| \leq \nu_{\text{czk}}(\lambda) + \nu_{\text{hide}}(\lambda) = \nu(\lambda).$$

□

Conclusion. After simulating proofs in CHHash and fixing the committed message to m' , the challenge view is independent of b ; any non-negligible advantage contradicts CZK or hiding.

J Proofs of Second DVS

J.1 Proof of Theorem 11 (Enhanced Simulation-Based Non-Transferability)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{DVS-ESimNonDeleg}}$: adversary gets $(\text{sk}_\Psi^s = w, \text{sk}_\Psi^v = \tau)$; $\mathcal{O}_{\text{LoR}}^b(m)$ returns a real proof if $b=0$ and a simulated proof if $b=1$.
- **Game 1** Answer LoR via the CZK challenger's unified oracle $\mathcal{O}_{P_b}^\tau((f, y, m), w)$ that outputs real/simulated depending on b .
- **Game 2** LoR always simulated: return $\text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m))$, independent of b .

Proof. Let S_i denote the event $\mathcal{A}$ outputs $b^* = b$ in Game i .

Game 0: Original experiment. Fig. 9. Sample

$$(\text{sk}_\Psi^s, \text{pk}_\Psi^s) = (w, (f, y = f(w))), \quad (\text{sk}_\Psi^v, \text{pk}_\Psi^v) = (\tau, \text{crs}_\Pi).$$

On LoR query m ,

$$\sigma \leftarrow \begin{cases} \text{Prf}_\Pi(\text{crs}_\Pi, (f, y, m), w), & b = 0, \\ \text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m)), & b = 1. \end{cases}$$

Let S_0 be success.

Game 1: LoR via CZK oracle. As in Game 0, but answer $\mathcal{O}_{\text{LoR}}^b(m)$ by forwarding $((f, y, m), w)$ to $\mathcal{O}_{P_b}^\tau$ and returning its output σ . Let S_1 be success.

Transition $0 \rightarrow 1$. By composable ZK, replacing locally produced real/simulated proofs with the CZK oracle is indistinguishable:

$$|\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{czk}}^{(1)}(\lambda).$$

$\mathcal{B}_{\text{czk}}^{(1)}(1^\lambda)$

1. Receive (crsn, τ) and access to $\mathcal{O}_{P_b}^\tau$ from the CZK challenger.
2. Choose $f \in \mathcal{F}$, sample w , set $y = f(w)$; give $(\text{sk}_\Psi^s, \text{sk}_\Psi^v) = (w, \tau)$ to $\mathcal{A}$ and publish $(\text{pk}_\Psi^s, \text{pk}_\Psi^v) = ((f, y), \text{crsn})$.
3. On LoR query m , return $\mathcal{O}_{P_b}^\tau((f, y, m), w)$.
4. Output b^* .

Advantage analysis.

$$|\Pr[S_0] - \tfrac{1}{2}| \leq |\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{czk}}(\lambda) = \nu(\lambda).$$

Therefore Ψ satisfies enhanced simulation-based non-transferability. $\square$

Conclusion. After answering LoR via the CZK oracle and then forcing LoR to output simulated proofs only, the view is b -independent even with $(\text{sk}_\Psi^s, \text{sk}_\Psi^v)$ known. Any non-negligible advantage would violate CZK.

K Proofs of Third CH

K.1 Proof of Theorem 17 (Full Collision Resistance)

Hop Overview.

- **Game 0** Original $\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-CollRes}}$: adversary outputs $(m^*, r^*, m'^*, r'^*, h^*)$ with $m^* \neq m'^*$, both checks accept, and *at least one* of (h^*, m^*) , (h^*, m'^*) is not from $\mathcal{O}_{\text{CHAdapt}}$.
- **Game 1** Bookkeeping: classify accepting tuples as external (E) vs. simulated (S) and deterministically turn E/S into E/E using a recorded extra external tuple for the same h^* .
- **Game 2** CRS switch: replace honest CRS by extractable CRS using CRS-indistinguishability; CHAdapt still via SIM_2 .
- **Game 3** Dual extraction on two external accepting tuples; any accept-but-fail breaks modified SSE.
- **Game 4** Binding hand-off with two extracted openings.

Proof. Let S_i be success in Game i . Let $\mathcal{Q} \subseteq \mathcal{H} \times \mathcal{M}$ record all (h, m) ever output by $\mathcal{O}_{\text{CHAdapt}}$; a tuple (h, m, r) is *external* iff $(h, m) \notin \mathcal{Q}$.

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \text{CH}}^{\text{F-CollRes}}(\lambda)$ with honestly sampled (crsn, ξ) and pp_Γ ; CHAdapt answered via $\text{SIM}_2(\text{crsn}, \tau, \cdot)$ where τ derives from ξ . $\mathcal{A}$ wins with $(m^*, r^*, m'^*, r'^*, h^*)$ such that both checks accept, $m^* \neq m'^*$, and at least one of (h^*, m^*) , (h^*, m'^*) is external. Let S_0 denote this event.

Game 1: Bookkeeping and E/S $\Rightarrow$ E/E. As in Game 0, but record every external accepting tuple in

$$\mathcal{R} \leftarrow \mathcal{R} \cup \{(h, m, r) \mid \text{CHCheck}(\text{pk}_{\text{ch}}, m, r, h) = 1 \wedge (h, m) \notin \mathcal{Q}\}.$$

If the final pair is E/S, replace the S-side by a distinct $(h^*, \tilde{m}, \tilde{r}) \in \mathcal{R}$ with $\tilde{m} \notin \{m^*, m'^*\}$, yielding an E/E pair for h^* . Let S_1 be success (same predicate after replacement).

Transition $0 \rightarrow 1$. Pure logging and deterministic post-processing:

$$|\Pr[S_1] - \Pr[S_0]| = 0.$$

Game 2: CRS switch (CRS-indistinguishability). Sample an extractable CRS

$$(\text{crs}_{\Pi}; \xi) \leftarrow \text{PG}_{\Pi}(1^\lambda, 1), \quad (\tau, \zeta) \leftarrow \text{SIM}_1(\xi),$$

and keep CHAdapt via $\text{SIM}_2(\text{crs}_{\Pi}, \tau, \cdot)$. Event S_2 .

Transition $1 \rightarrow 2$. By CRS-indistinguishability of Π ,

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{crsind}}(\lambda).$$

- $\mathcal{B}_{\text{crsind}}(1^\lambda)$
1. Receive either an honest or an extractable crs_{Π} (hidden bit) from the CRS-indistinguishability challenger; obtain τ (and, if provided, ζ) via its interface.
 2. Sample honest pp_{r} , set $\text{pk}_{\text{ch}} = (\text{crs}_{\Pi}, \text{pp}_{\text{r}})$, answer CHAdapt via $\text{SIM}_2(\text{crs}_{\Pi}, \tau, \cdot)$, and run $\mathcal{A}$ implementing Game 1 logic.
 3. Output $\mathcal{A}$'s success bit; this distinguishes the two setups with advantage $|\Pr[S_2] - \Pr[S_1]|$.

Game 3: Dual extraction on external tuples. After enforcing E/E, run

$$w_1 \leftarrow \mathcal{E}(\text{crs}_{\Pi}, \zeta, (\text{pp}_{\text{r}}, h^*, m^*), r^*), \quad w_2 \leftarrow \mathcal{E}(\text{crs}_{\Pi}, \zeta, (\text{pp}_{\text{r}}, h^*, m'^*), r'^*).$$

If Verify accepts but any *external* extraction fails, charge modified-SSE. Let S_3 be “both extractions succeed”.

Transition $2 \rightarrow 3$. By modified SSE and a union bound over two extractions,

$$|\Pr[S_3] - \Pr[S_2]| \leq 2\nu_{\text{msse}}^{(2)}(\lambda).$$

- $\mathcal{B}_{\text{msse}}^{(2)}(1^\lambda)$
1. Receive *only* crs_{Π} and access to the simulator oracle $\text{SIM}_2(\text{crs}_{\Pi}, \cdot)$ from the mSSE challenger; maintain $\mathcal{Q}, \mathcal{R}$.
 2. Answer CHAdapt via the simulator and log all $(h, m) \in \mathcal{Q}$.
 3. Run $\mathcal{A}$ with the Game 1 E/S $\rightarrow$ E/E rule; obtain two *external* accepting pairs $(x_1, \pi_1), (x_2, \pi_2) \notin \mathcal{Q}$ for the same h^* .
 4. If extraction fails on at least one external pair, choose uniformly one failing pair (x_b, π_b) and return it. The 1/2 selection yields the global factor 2.

Game 4: Binding hand-off. On S_3 , output $(h^*, m^*, w_1, m'^*, w_2)$ to the binding challenger. Let S_4 denote success there.

Transition $3 \rightarrow 4$. Pure relabeling: $|\Pr[S_4] - \Pr[S_3]| = 0$.

- $\mathcal{B}_{\text{bind}}(1^\lambda)$
1. Receive pp_{r} ; generate $(\text{crs}_{\Pi}, \xi) \leftarrow \text{PG}_{\Pi}(1^\lambda, 1)$ and $(\tau, \zeta) \leftarrow \text{SIM}_1(\xi)$.
 2. Run $\mathcal{A}$ with $\text{pk}_{\text{ch}} = (\text{crs}_{\Pi}, \text{pp}_{\text{r}})$; answer CHAdapt via SIM_2 ; keep $\mathcal{Q}, \mathcal{R}$ and enforce E/S $\rightarrow$ E/E.
 3. Extract w_1, w_2 as above and return two distinct openings for h^* .

Advantage analysis. From the transitions,

$$\Pr[S_0] \leq \nu_{\text{crsind}}(\lambda) + 2\nu_{\text{msse}}^{(2)}(\lambda) + \Pr[S_4].$$

Binding succeeds whenever S_4 occurs, hence

$$\Pr[S_0] \leq \nu_{\text{crsind}}(\lambda) + 2\nu_{\text{msse}}^{(2)}(\lambda) + \nu_{\text{bind}}(\lambda).$$

□

Conclusion. With bookkeeping that reduces E/S to E/E, any successful collision either yields an accepting external proof without an extractable witness (mSSE break) or two distinct openings for the same h^* (binding break). Thus CH'' is fully collision-resistant.

K.2 Proof of Theorem 18 (Full Indistinguishability)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{F-Ind}}$.
- **Game 1** Simulate the proof inside CHHash using the NIZK trapdoor.
- **Game 2** In the challenge oracle, always commit to m' ; proofs remain simulated.

Proof. Let S_i be the event $\mathcal{A}$ outputs $b^* = b$ in Game i .

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \text{CH}''}^{\text{F-Ind}}(1^\lambda)$. Let S_0 denote success.

Game 1: Simulated proofs in CHHash. As in Game 0, but generate π in CHHash using the simulation trapdoor $\tau \leftarrow \text{SIM}_1(\xi)$. Let S_1 denote success.

Transition 0 $\rightarrow$ 1. By trapdoor/composable ZK of Π ,

$$|\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{tdzk}}(\lambda).$$

- $\mathcal{B}^{\text{tdzk}}(1^\lambda)$

 1. Receive $(\text{crs}_\Pi; \xi)$ with $\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) = 0$ from the TDZK challenger.
 2. Implement the challenge oracle by committing as in the scheme but producing the CHHash proof via the TDZK prover/simulator interface tied to ξ . 8
 3. Run $\mathcal{A}$ and return its output bit b^* to the challenger.

Game 2: Fix commitment to m' . As in Game 1, but the challenge oracle always commits to m' independent of b ; keep simulated proofs. Let S_2 denote success.

Transition 1 $\rightarrow$ 2. By adversarial hiding of Γ ,

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{advhide}}(\lambda).$$

- $\mathcal{B}^{\text{advhide}}(1^\lambda)$

 1. Receive pp_Γ and access to the hiding challenger $\mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(\cdot, \cdot)$.
 2. Obtain $(\text{crs}_\Pi, \tau, \zeta)$. Set $\text{pk}_{\text{ch}} = (\text{crs}_\Pi, \text{pp}_\Gamma)$.
 3. On a challenge (m, m') from $\mathcal{A}$: query $C \leftarrow \mathcal{O}_{\text{Cmt}_\Gamma}^{\text{hid}}(m, m')$ and return (C, π) with $\pi \leftarrow \text{SIM}_2(\text{crs}_\Pi, \tau, (\text{pp}_\Gamma, C, m'))$.
 4. Use $\mathcal{A}$'s bit to answer the hiding challenger.

Advantage analysis. In Game 2, outputs are b -independent, so $\Pr[S_2] = \frac{1}{2}$. Hence

$$|\Pr[S_0] - \frac{1}{2}| \leq \nu_{\text{tdzk}}(\lambda) + \nu_{\text{advhide}}(\lambda).$$

□

Conclusion. After simulating the proof in CHHash and fixing the committed message in the challenge oracle to m' , the view is independent of b . Any non-negligible advantage would contradict NIZK ZK or commitment hiding. Therefore CH'' satisfies Full-Indistinguishability.

L Proofs of Third DVS

L.1 Proof of Theorem 19 (Unforgeability)

Hop Overview.

- **Game 0** Original $\text{Exp}^{\text{DVS-Unforge}}_{\mathcal{A}, \Psi''}(\lambda)$.
- **Game 1** Switch to an extractable CRS $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; answering oracles unchanged.
- **Game 2** Run $w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, (f, y, m^*), \sigma^*)$ on any accepting forgery.
- **Game 3** If $f(w^*) = y$, output w^* to invert f ; else charge SSE.

Proof. Let S_i denote success in Game i .

Game 0: Original experiment. $\text{Exp}^{\text{DVS-Unforge}}_{\mathcal{A}, \Psi''}(\lambda)$. S_0 : $\mathcal{A}$ outputs a fresh (m^*, σ^*) with $\text{Verify}_\Psi(\text{pk}_\Psi^s, m^*, \sigma^*) = 1$.

Game 1: Extractable CRS. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; publish $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$; answer Sign_Ψ as in the scheme (when the scheme calls for simulation, use SIM_2 with τ ; otherwise prove honestly). Let S_1 be success here.

Transition 0 $\rightarrow$ 1. The marginal distribution of (crs_Π, τ) in $\mathcal{E}_1$ matches SIM_1 (scheme setup), so $|\Pr[S_1] - \Pr[S_0]| = 0$.

Game 2: Extraction. On a valid forgery for $x^* = (f, y, m^*)$ run

$$w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, x^*, \sigma^*).$$

Let S_2 be “forge and extraction succeeds”.

Transition 1 $\rightarrow$ 2. If $\text{Verify}_\Psi = 1$ yet extraction fails on a fresh proof, SSE is broken:

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{sse}}(\lambda).$$

Game 3: Inversion check. Let S_3 be “forge, extraction succeeds, and $f(w^*) = y$ ”.

Transition 2 $\rightarrow$ 3. If verify accepts and extraction succeeds but $f(w^*) \neq y$, the accepting proof lacks a valid witness, contradicting SSE:

$$|\Pr[S_3] - \Pr[S_2]| \leq \nu_{\text{sse}}(\lambda).$$

Reductions.

$\mathcal{B}_{\text{sse}}(1^\lambda)$ 1. Receive *only* crs_Π from the SSE challenger; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$.
 2. Choose $f \in \mathcal{F}$, sample $w \leftarrow C_\lambda$, set $y \leftarrow f(w)$ and publish $\text{pk}_\Psi^s \leftarrow (f, y)$.
 3. Answer all signing queries on m by producing a *real* proof $\pi \leftarrow \text{Prf}_\Pi(\text{crs}_\Pi, (f, y, m), w)$; return σ and record it in $\mathcal{Q}_{\text{sign}}$.
 4. Run $\mathcal{A}$; on a fresh accepting forgery (m^*, σ^*) with NIZK part π^* for $x^* = (f, y, m^*)$, output (x^*, π^*) to the SSE challenger.

$\mathcal{B}_{\text{owf}}(f, y)$ 1. Get OWF challenge (f, y) .
 2. Sample $(\text{crs}_\Pi, \tau, \zeta) \leftarrow \mathcal{E}_1(1^\lambda)$; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$, $\text{pk}_\Psi^s \leftarrow (f, y)$.
 3. Answer signing via $\text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m))$ (as specified by the scheme).
 4. Run $\mathcal{A}$; on forgery (m^*, σ^*) compute $w^* \leftarrow \mathcal{E}_2(\text{crs}_\Pi, \zeta, (f, y, m^*), \sigma^*)$ and output w^* .

Advantage analysis.

$$\Pr[S_0] \leq 0 + \nu_{\text{sse}} + \nu_{\text{sse}} + \nu_{\text{owf}}.$$

Thus $\Pr[S_0] \leq 2\nu_{\text{sse}}(\lambda) + \nu_{\text{owf}}(\lambda)$. $\square$

Conclusion. Either an accepting fresh proof is non-extractable (SSE break) or extraction yields w^* with $f(w^*) = y$ (OWF break). Hence Ψ'' is strongly unforgeable.

L.2 Proof of Theorem 20 (Simulation-Based Non-Transferability)

Hop Overview.

- **Game 0** Original $\text{Exp}_{\mathcal{A}, \Psi''}^{\text{DVS-FSimNonTransfer}}$ with LoR oracle outputting honest vs. simulated depending on b .
- **Game 1** Replace LoR answers by TDZK challenger's oracle (real/simulated internally).
- **Game 2** Force LoR to always return simulated proofs; view becomes b -independent.

Proof. Let S_i be the event $b^* = b$ in Game i .

Game 0: Original experiment. $\text{Exp}_{\mathcal{A}, \Psi''}^{\text{DVS-FSimNonTransfer}}(\lambda)$. Let S_0 denote success.

Game 1: LoR via TDZK oracle. Answer $\mathcal{O}_{\text{LoR}}^b(m)$ by forwarding $((f, y, m), w)$ to the TDZK unified oracle $\mathcal{O}_{P_b}^\tau$ that returns a real proof for $b=0$ or a simulated proof for $b=1$. Let S_1 denote success.

Transition 0 $\rightarrow$ 1. By trapdoor-detectable ZK,

$$|\Pr[S_1] - \Pr[S_0]| \leq \nu_{\text{tdzk}}^{(1)}(\lambda).$$

$\mathcal{B}_{\text{tdzk}}^{(1)}(1^\lambda)$ 1. Receive $(\text{crs}_\Pi; \xi)$ and access to $\mathcal{O}_{P_b}^\tau$ from the TDZK challenger; set $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$.
 2. Choose $f \in \mathcal{F}$, sample $w \leftarrow C_\lambda$, set $y \leftarrow f(w)$ and $\text{pk}_\Psi^s \leftarrow (f, y)$; give $(\text{pk}_\Psi^s, \text{pk}_\Psi^v)$ to $\mathcal{A}$.
 3. On LoR query m , forward $((f, y, m), w)$ to $\mathcal{O}_{P_b}^\tau$ and return its output.
 4. Output $\mathcal{A}$'s bit to the challenger.

Game 2: Always simulated LoR. Make $\mathcal{O}_{\text{LoR}}$ ignore b and always output $\text{SIM}_2(\text{crs}_\Pi, \tau, (f, y, m))$. Let S_2 denote success.

Transition 1 $\rightarrow$ 2. Real vs. simulated is indistinguishable under composition:

$$|\Pr[S_2] - \Pr[S_1]| \leq \nu_{\text{tdzk}}^{(2)}(\lambda).$$

$\underline{B_{\text{tdzk}}^{(2)}(1^\lambda)}$

1. Receive $(\text{crs}_\Pi; \xi)$ and access to TDZK interfaces producing real and simulated proofs for the same statements.
2. Publish $\text{pk}_\Psi^v \leftarrow \text{crs}_\Pi$, choose f, w , set $y \leftarrow f(w)$, $\text{pk}_\Psi^s \leftarrow (f, y)$; run $\mathcal{A}$.
3. Give $\mathcal{A}$ access to either $\mathcal{O}_{\text{LoR}}^{(1)}$ (uses $\mathcal{O}_{P_b}^r$, as in Game 1) or $\mathcal{O}_{\text{LoR}}^{(2)}$ (always simulated, as in Game 2).
4. Use $\mathcal{A}$'s distinguishing advantage to distinguish real vs. simulated in TDZK.

Advantage analysis. Game 2 is b -independent, so $\Pr[S_2] = \frac{1}{2}$ and

$$\left| \Pr[S_0] - \frac{1}{2} \right| \leq \nu_{\text{tdzk}}^{(1)}(\lambda) + \nu_{\text{tdzk}}^{(2)}(\lambda) = \nu(\lambda).$$

□

Conclusion. Routing LoR through the TDZK oracle and then fixing it to simulated outputs removes any b -dependence. Any non-negligible advantage would violate TDZK; therefore Ψ'' satisfies full simulation-based non-transferability.

M Proofs of TD-NZ Instantiation

M.1 Proof of Theorem 12 (Simulation-Based Non-Transferability)

Hop Overview.

- **Game 0** Real completeness.
- **Game 1** PRF-uniformization.
- **Game 2** Branch-collision accounting.

Proof. Fix the bilinear group and the random oracles. Let S_i denote “verifier rejects a true statement” in Game i .

Game 0 (Real). CRS via $\text{PG}_\Pi(1^\lambda, b)$; for $R(x, w) = 1$ check $\forall \text{fy}_\Pi(\text{crs}_\Pi, x, \text{Prf}_\Pi(\text{crs}_\Pi, x, w))$. Proving and verification are Groth’s; for any fixed CRS completeness is perfect. Rejection can only arise from a degenerate CRS.

Game 1 (PRF-uniformized). Run as G_0 but use $x \leftarrow U_1$, $y \leftarrow U_2$, $r_u \leftarrow U_3$, $s_v \leftarrow U_4$ with $U_i \leftarrow_{\$} \mathbb{Z}_p$. Let S_1 be rejection in G_1 .

Transition $0 \rightarrow 1$. Replace $\text{PRFEval}(\xi', i)$ by $U_i \leftarrow_{\$} \mathbb{Z}_p$ for $i \in \{1, 2, 3, 4\}$. Then $|\Pr[S_1] - \Pr[S_0]| \leq \mathcal{A}_{\text{PRF}}(\lambda)$.

$\underline{B_{\text{prf}}(1^\lambda)}$

1. Access oracle $\mathcal{O}$ that is either $\text{PRF}(\xi', \cdot)$ or uniform over $\mathbb{Z}_p$.
2. Run G_0 verbatim *except* set

$$x \leftarrow \mathcal{O}(1), \quad y \leftarrow \mathcal{O}(2), \quad r_u \leftarrow \mathcal{O}(3), \quad s_v \leftarrow \mathcal{O}(4).$$

3. Output whatever $\mathcal{A}$ outputs.

Game 2 (Collision-accounted). Same as G_1 , with the above collision treated as a bad event. Let S_2 be rejection in G_2 .

Transition 1 $\rightarrow$ 2. Let $(f^{(0)}, h^{(0)}) = (\mathcal{H}_{\mathbb{G}_1}(1, \xi'), \mathcal{H}_{\mathbb{G}_1}(2, \xi'))$ and $(f^{(1)}, h^{(1)}) = (g_1^X, g_1^Y)$ with $X, Y \leftarrow_{\S} \mathbb{Z}_p$. In the ROM these are independent uniforms in $\mathbb{G}_1$, hence

$$\Pr[(f^{(0)}, h^{(0)}) = (f^{(1)}, h^{(1)})] = \frac{1}{|\mathbb{G}_1|^2} \leq \frac{1}{p^2},$$

and thus $|\Pr[S_2] - \Pr[S_1]| \leq \frac{1}{p^2}$.

- | | |
|------------------------------------|--|
| <u>C_{rom}</u> | 1. Treat $\mathcal{H}_{\mathbb{G}_1}(1, \cdot)$ and $\mathcal{H}_{\mathbb{G}_1}(2, \cdot)$ as independent ROs to $\mathbb{G}_1$ (domain separation).
2. Note g_1^X, g_1^Y with $X, Y \leftarrow_{\S} \mathbb{Z}_p$ are uniform in $\mathbb{G}_1$ and independent of the ROs.
3. Conclude $\Pr[(f^{(0)}, h^{(0)}) = (f^{(1)}, h^{(1)})] = 1/ \mathbb{G}_1 ^2$; no assumption needed. |
|------------------------------------|--|

Analysis. By triangle inequality,

$$\Pr[\text{reject on } R(x, w)=1] \leq \mathcal{A}_{\text{PRF}}(\lambda) + \frac{1}{p^2} = \nu(\lambda).$$

Conclusion. Outside a negligible CRS branch-collision, Groth's completeness applies unchanged. The scheme is correct with negligible error. $\square$

M.2 Proof of Theorem 13 (CRS-Indistinguishability)

Hop Overview.

- **Game 0** Real $b=0$: $(\text{crs}_{\Pi}; \xi) \leftarrow \text{PG}_{\Pi}(1^\lambda, 0)$; $(\tau, \perp) \leftarrow \text{SIM}_1(\xi, 0)$; output (crs_{Π}, τ) .
- **Game 1** Uniformize r_u, s_v : replace $\text{PRFEval}(\xi', 3), \text{PRFEval}(\xi', 4)$ by $U_3, U_4 \leftarrow_{\S} \mathbb{Z}_p$.
- **Game 2** Replace (f, h) : set $(f, h) \leftarrow (g_1^X, g_1^Y)$ with $X, Y \leftarrow_{\S} \mathbb{Z}_p$ (ROM counting).
- **Game 3** Real $b=1$: $(\text{crs}_{\Pi}; \xi) \leftarrow \text{PG}_{\Pi}(1^\lambda, 1)$; $(\tau, \zeta) \leftarrow \text{SIM}_1(\xi, 1)$; output (crs_{Π}, τ) .

Proof. Fix the bilinear group and random oracles. Let S_i be the event “ $\mathcal{A}$ outputs 1 on (crs_{Π}, τ) ” in Game i .

Game 0 (Real, $b=0$). Here $f = \mathcal{H}_{\mathbb{G}_1}(1, \xi')$, $h = \mathcal{H}_{\mathbb{G}_1}(2, \xi')$, and $(r_u, s_v) = (\text{PRFEval}(\xi', 3), \text{PRFEval}(\xi', 4))$. $\tau = (r_u, s_v, \text{sk}_{\Sigma})$ is computed by $\text{SIM}_1(\xi, 0)$ from ξ' ; sk_{Σ} is derived deterministically from ξ' .

Game 1 (Uniform r_u, s_v). Same as G_0 but with $(r_u, s_v) = (U_3, U_4)$. Let S_1 be the success event.

Transition 0 $\rightarrow$ 1. Replace (r_u, s_v) by independent uniforms $(U_3, U_4) \leftarrow_{\S} \mathbb{Z}_p^2$. Then

$$|\Pr[S_1] - \Pr[S_0]| \leq \mathcal{A}_{\text{PRF}}^{(3,4)}(\lambda).$$

- | | |
|---|--|
| <u>$B_{\text{prf}}^{(3,4)}(1^\lambda)$</u> | 1. Access oracle $\mathcal{O}$ that on input $i \in \{3, 4\}$ returns either $\text{PRF}(\xi', i)$ or uniform in $\mathbb{Z}_p$.
2. Run G_0 but set $r_u \leftarrow \mathcal{O}(3)$, $s_v \leftarrow \mathcal{O}(4)$; keep all other values as in G_0 .
3. Output $\mathcal{A}(\text{crs}_{\Pi}, \tau)$.
<i>Advantage.</i> If $\mathcal{O}$ is PRF then the view is G_0 ; if uniform then G_1 . Thus $ \Pr[S_1] - \Pr[S_0] \leq \mathcal{A}_{\text{PRF}}^{(3,4)}(\lambda)$. |
|---|--|

Game 2 (Uniformized (f, h, r_u, s_v)). Now $(f, h) = (g_1^X, g_1^Y)$ with $X, Y \leftarrow_{\$} \mathbb{Z}_p$ and $(r_u, s_v) = (U_3, U_4) \leftarrow_{\$} \mathbb{Z}_p^2$. $\tau = (r_u, s_v, \text{sk}_{\Sigma})$ as before.

Transition $1 \rightarrow 2$. Replace $(f, h) = (\mathcal{H}_{G_1}(1, \xi'), \mathcal{H}_{G_1}(2, \xi'))$ by (g_1^X, g_1^Y) with $X, Y \leftarrow_{\$} \mathbb{Z}_p$. In the ROM, both pairs are independent uniform in $\mathbb{G}_1^2$. Hence

$$|\Pr[S_2] - \Pr[S_1]| = 0.$$

C_{rom} 1. $\mathcal{H}_{G_1}(1, \cdot)$ and $\mathcal{H}_{G_1}(2, \cdot)$ are independent ROs to G_1 (domain separation).
 2. g_1^X, g_1^Y with $X, Y \leftarrow_{\$} \mathbb{Z}_p$ are independent uniform in $\mathbb{G}_1$ and independent of the ROs.
 3. Therefore the joint distributions of (f, h) in G_1 and G_2 are identical.

Game 3 (Real, $b=1$). This is exactly the distribution in the right-hand side of the CRS-Indistinguishability game: $(\text{crs}_{\Pi}; \xi) \leftarrow \text{PG}_{\Pi}(1^\lambda, 1)$, $(\tau, \zeta) \leftarrow \text{SIM}_1(\xi, 1)$, and output (crs_{Π}, τ) .

Transition $2 \rightarrow 3$. Move from uniforms back to PRF-derived values for all four tags $(1, 2, 3, 4)$, matching $b=1$. Then

$$|\Pr[S_3] - \Pr[S_2]| \leq \mathcal{A}_{\text{PRF}}^{(1,2,3,4)}(\lambda).$$

$\mathcal{B}_{\text{prf}}^{(1,2,3,4)}(1^\lambda)$ 1. Access oracle $\mathcal{O}$ for tags $i \in \{1, 2, 3, 4\}$ returning either $\text{PRF}(\xi', i)$ or uniform in $\mathbb{Z}_p$.
 2. Run G_2 but set $X \leftarrow \mathcal{O}(1)$, $Y \leftarrow \mathcal{O}(2)$, $r_u \leftarrow \mathcal{O}(3)$, $s_v \leftarrow \mathcal{O}(4)$; set $f \leftarrow g_1^X$, $h \leftarrow g_1^Y$.
 3. Output $\mathcal{A}(\text{crs}_{\Pi}, \tau)$.
Advantage. If $\mathcal{O}$ is PRF the view is G_3 (real $b=1$); if uniform the view is G_2 . Hence $|\Pr[S_3] - \Pr[S_2]| \leq \mathcal{A}_{\text{PRF}}^{(1,2,3,4)}(\lambda)$.

Analysis. By the triangle inequality,

$$|\Pr[S_3] - \Pr[S_0]| \leq \underbrace{\mathcal{A}_{\text{PRF}}^{(3,4)}(\lambda)}_{\text{Transition } 0 \rightarrow 1} + \underbrace{0}_{1 \rightarrow 2} + \underbrace{\mathcal{A}_{\text{PRF}}^{(1,2,3,4)}(\lambda)}_{\text{Transition } 2 \rightarrow 3} = \nu(\lambda).$$

This matches the advantage notion in the CRS-Indistinguishability definition, since G_0 and G_3 instantiate its left and right experiments.

Conclusion. (crs_{Π}, τ) under $b = 0$ and $b = 1$ are computationally indistinguishable. Therefore Π satisfies CRS-Indistinguishability. $\square$

M.3 Proof of Theorem 14 (Trapdoor-Detectable Zero-Knowledge)

Hop Overview.

- **Game 0** TD-ZK experiment of Fig. 10 with branch $b=0$.
- **Game 1.** t For $t \in \{0, \dots, q\}$: on admissible queries, the first t answers are simulated with $\tau \leftarrow \text{SIM}_1(\xi, 0)$; later admissible answers are real. Note $G_{1,0} \equiv G_0$.
- **Game 2** Fully simulated answers on all admissible queries ($G_{1,q} \equiv G_2$); matches branch $b=1$ of Fig. 10.

Proof. Fix the bilinear group and the random oracles. Let q be the polynomially bounded number of *admissible* oracle queries $(\text{crs}_\Pi, \xi, x, w)$ with $R(x, w) = 1$ and $\text{DtctTd}_\Pi(\text{crs}_\Pi, \xi) = 0$. Let S_i denote “ $\mathcal{A}$ outputs 1 in Game i .”

Game 0 (Real, $b=0$). Oracle returns $\text{Prf}_\Pi(\text{crs}_\Pi, x, w)$ on admissible queries and $\perp$ otherwise.

Transition 0 $\rightarrow$ 1.0. Pure renaming: $G_{1.0} \equiv G_0$, hence $|\Pr[S_{1.0}] - \Pr[S_0]| = 0$.

Game 1. t to Game 1. $(t+1)$ ($t = 0, \dots, q-1$). Swap the $(t+1)$ -st admissible answer from real to simulated using the same $\tau \leftarrow \text{SIM}_1(\xi, 0)$.

Transition 1. $t \rightarrow$ 1. $(t+1)$. Swap the $(t+1)$ -st admissible answer from real to simulated using the same $\tau \leftarrow \text{SIM}_1(\xi, 0)$. By the generic ZK reduction $\mathcal{B}_{\text{ZK}}$,

$$|\Pr[S_{1.(t+1)}] - \Pr[S_{1.t}]| \leq \mathcal{A}_{\text{ZK}}(\lambda).$$

$\mathcal{B}_{\text{ZK}}(1^\lambda)$ 1. Access a ZK challenge oracle $\mathcal{O}_{\text{ZK}}$ that on $(\text{crs}_\Pi, x, w, \tau)$ returns either $\text{Prf}_\Pi(\text{crs}_\Pi, x, w)$ or $\text{SIM}_2(\text{crs}_\Pi, \tau, x)$ under a hidden bit.
 2. Emulate $G_{1.t}$ for $\mathcal{A}$: for each oracle query $(\text{crs}_\Pi, \xi, x, w)$, return $\perp$ if not admissible; for the first t admissible queries, answer exactly as in $G_{1.t}$; on the $(t+1)$ -st admissible query, compute $\tau \leftarrow \text{SIM}_1(\xi, 0)$ and answer via $\mathcal{O}_{\text{ZK}}(\text{crs}_\Pi, x, w, \tau)$; answer later admissible queries real.
 3. Output the bit of $\mathcal{A}$.
Advantage. If $\mathcal{O}_{\text{ZK}}$ serves real proofs the view is $G_{1.t}$; if simulated the view is $G_{1.(t+1)}$. Thus $|\Pr[S_{1.(t+1)}] - \Pr[S_{1.t}]| \leq \mathcal{A}_{\text{ZK}}(\lambda)$.

Transition 1. $q \rightarrow$ 2. By definition $G_{1.q} \equiv G_2$ (all admissible answers simulated), so $|\Pr[S_2] - \Pr[S_{1.q}]| = 0$.

Analysis. Telescoping over q swaps:

$$|\Pr[S_2] - \Pr[S_0]| \leq \sum_{t=0}^{q-1} |\Pr[S_{1.(t+1)}] - \Pr[S_{1.t}]| \leq q \cdot \mathcal{A}_{\text{ZK}}(\lambda) = \nu(\lambda).$$

Thus $\left| \Pr[\mathbf{Exp}^{\text{TD-ZK}}=1] - \frac{1}{2} \right| \leq \nu(\lambda)$.

Conclusion. The scheme satisfies trapdoor-detectable zero-knowledge w.r.t. Fig. 10. $\square$

N On the Usage of AI

As required by recent AI Tool Policies (e.g., ACM guidance adopted by several IACR venues), we disclose *what* AI tools were used in preparing this manuscript, *how* they were used, and *how we ensured* the integrity of the work. We (the authors) accept full responsibility for all content.

AI-Tool Disclosure. We used a generative AI assistant (OpenAI ChatGPT, model GPT-5 Thinking) for language polishing and $\text{\LaTeX}$ boilerplate. Scientific ideas, definitions, theorems, and proofs—including TD-ZK, the SSE variants, and all game hops—are by the authors. All AI-assisted output was reviewed and edited by the authors. No automated proof tools, code execution, or web search via the assistant were used.

Summary. We used the assistant as a virtual reviewer and writing aide. It helped with phrasing, consistency of notation, and L^AT_EX layout (figures/experiments). All technical content is authored, verified, and owned by us; the assistant did not generate ideas, proofs, or claims.

N.1 Tools, Versions, and Access Modality

- **Tool:** OpenAI ChatGPT (model: GPT-5 Thinking)
- **Scope of use:** Interactive drafting and editing of L^AT_EX prose; wording and stylistic consistency; L^AT_EX boilerplate for experiments (e.g., `tabbing` blocks for TD-ZK and MSimSoundExt), figures, and environments; red-teaming language for definitions and security games; notation consistency ($\mathcal{A}, \lambda, \mathcal{Q}$).
- **Non-use:** No code execution, no empirical evaluation, no automated proof assistants, and no external web browsing via the tool for scientific claims. All citations and attributions were curated and verified by the authors.

N.2 Summary of AI Involvement (High Level)

The assistant was used *as a writing and presentation aid* to accelerate drafting of sections whose scientific content we designed. We supplied the mathematical intent, definitions, experiments, game hops, and security goals. The assistant proposed wording and L^AT_EX snippets. We revised and validated every passage before inclusion.

N.3 Granular Disclosure by Artifact

Expository prose. First-pass phrasing and polishing of motivation, overviews, and “story-of-the-proof” text for TD-ZK and SSE sections; we verified every sentence for correctness and intent.

Formal definitions and experiments. We specified all interfaces and predicates. The assistant helped render experiments in a consistent L^AT_EX style and harmonize names ($\mathcal{A}, \lambda, \mathcal{Q}$, oracle labels). All semantics were authored and checked by us.

Construction sections. For the TD-ZK instantiation and the modified SSE experiment, we provided the constructions and conditions; the assistant aided with layout and wording only.

Figures and pseudo-code. The assistant produced L^AT_EX layout (`adjustbox`, `tabbing`, captions). We checked inputs/outputs, abort conditions, and filters (e.g., admissibility via $\text{DtctTd}_{\Pi}(\text{crs}_{\Pi}, \xi) = 0$).

Consistency passes. We used the assistant to flag potential inconsistencies (quantifier order, replay filters, use of $\perp$, parameter dependence). We decided on fixes and edited the text.

Negative use-cases. No AI-generated proof steps were accepted; no auto-citations; no literature search. All references and claims were inserted and verified by the authors.

N.4 What the AI Did *Not* Do

- Did not originate research questions, notions, or constructions.
- Did not perform proofs or generate technical claims.
- Did not search related work or browse the web.
- Did not run experiments, verify code, or produce empirical results.

N.5 Quality Control and Risk Mitigation

1. **Manual proof audit.** Every AI-assisted definition/experiment/hop description was checked by at least one author; each transition has an explicit assumption/reduction and named success event.
2. **Assumption hygiene.** Uses of ZK and SSE are tied to specific hops; no circular justifications.

N.6 Provenance, Reproducibility, and Transparency

- **Granularity of disclosure.** AI materially influenced presentation in: captions of TD-ZK and MSimSoundExt figures, expository narrative in the construction section, and security-game explanations. Final text reflects authors’ edits.
- **Attribution and authorship.** The assistant is not an author, provided no originality, and bears no responsibility.

N.7 Data, Ethics, and Privacy

- **No personal data.** We did not provide personal or sensitive data to the assistant.
- **No datasets.** The paper is theoretical; no datasets were used or generated.
- **Plagiarism avoidance.** AI output was treated as writing assistance; we ensured originality of expression for our contributions. All citations are authored.

N.8 Limitations of AI Assistance

AI drafts can miss edge cases or alter scoping. Our audits targeted these risks (e.g., admissibility guards, oracle domains, success predicates). Any remaining errors are ours.

N.9 Policy-Facing Statement

We used a generative AI assistant (OpenAI ChatGPT, model “GPT-5 Thinking”) for language polishing, L^AT_EX boilerplate, and drafting of expository text and experiment layouts under author supervision. The scientific ideas, formal definitions, theorems, and proofs are the authors’ work. We verified every AI-assisted passage for correctness and clarity, incorporated material where appropriate, and accept full responsibility for the content.

Representative prompts (examples).

1. **AI-usage disclosure paragraph (be honest).**
Draft a 120–160 word “AI Tool Disclosure” paragraph for an LNCS paper on trapdoor-detectable zero-knowledge. Be explicit that the assistant helped only with language polishing and $\text{\LaTeX}$ boilerplate, not with ideas, proofs, experiments, or citations. State that this paragraph itself was drafted by the assistant at the authors’ request, and that the authors reviewed, edited, and accept full responsibility. Neutral academic tone; no hype; no policy advice.
2. **Language polish without content changes.**
Polish the following paragraph for clarity and concision. Do not change any technical claims, symbols, or notation. Keep math and identifiers verbatim. Output only the revised paragraph.
Text: <<<! $\text{\LaTeX}$ PARAGRAPH!>>>
3. **$\text{\LaTeX}$ boilerplate for side-by-side experiments.**
Produce $\text{\LaTeX}$ for two top-aligned boxes (`minipage + adjustbox`) placed side by side, compatible with `llncs` (no `caption/subcaption`).
Left box title: <<<! $\text{\LaTeX}$ LEFT TITLE!>>>; right box title: <<<! $\text{\LaTeX}$ RIGHT TITLE!>>>.
Each box contains a compact `tabbing` block. Thin frame, tight spacing, no vertical stretch.
4. **Consistency review (virtual reviewer).**
Review the excerpt only for consistency: quantifier order, notation ($\mathcal{A}$, λ , $\mathcal{Q}$ sets), oracle interfaces, and replay-filter conditions. Report issues as a bullet list with one-line justifications. No rewrites, no new content.
Excerpt: <<<! $\text{\LaTeX}$ TEXT!>>>
5. **Red-team the security experiment.**
Stress-test this experiment for likely gaps: missing abort conditions, insufficient replay filter, unbound handles, and edge cases (unregistered keys, cross-context queries). Return a checklist (Yes/No + one sentence) and point to exact lines.
Experiment: <<<! $\text{\LaTeX}$ CODE!>>>

N.10 Why We Included This Section

To distinguish presentation-level assistance from idea/proof authorship and to state the checks we performed so that trust in the claims stems from transparent process and verifiable reasoning rather than tooling.